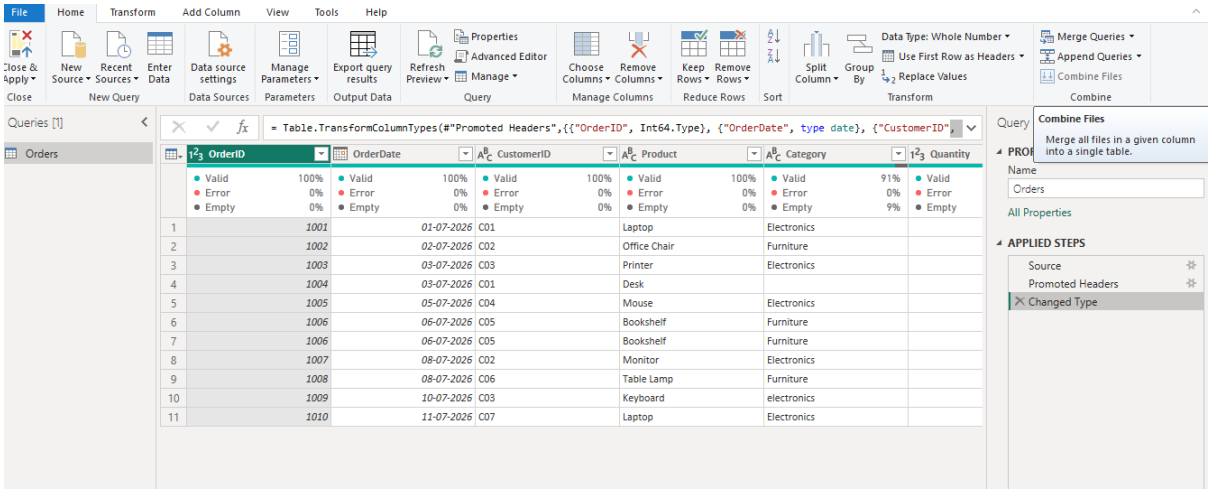
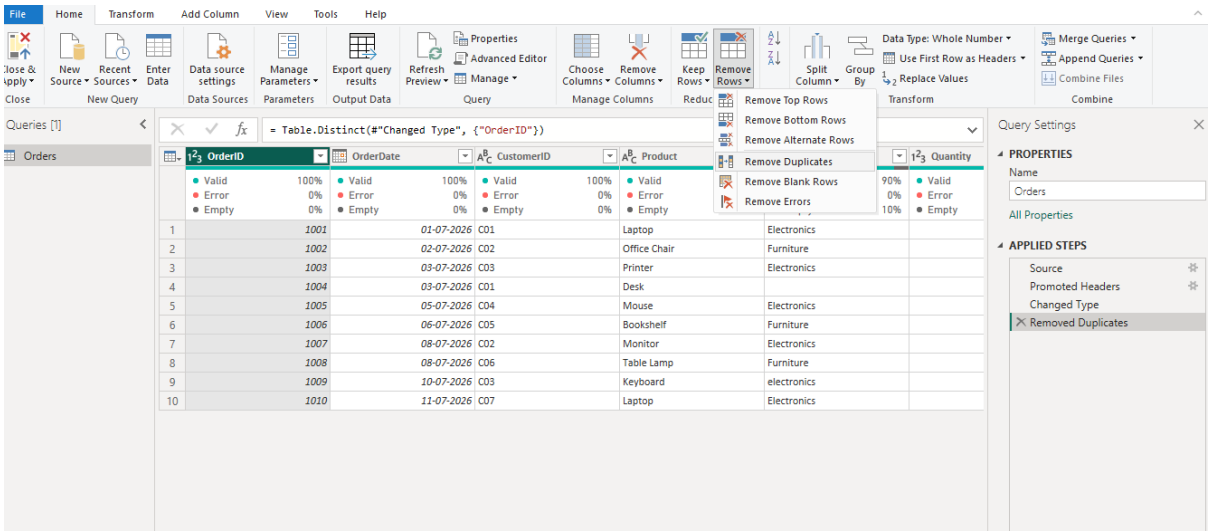
PowerBI- https://github.com/Geetanjali-Bhardwaj/RollNo_LA1

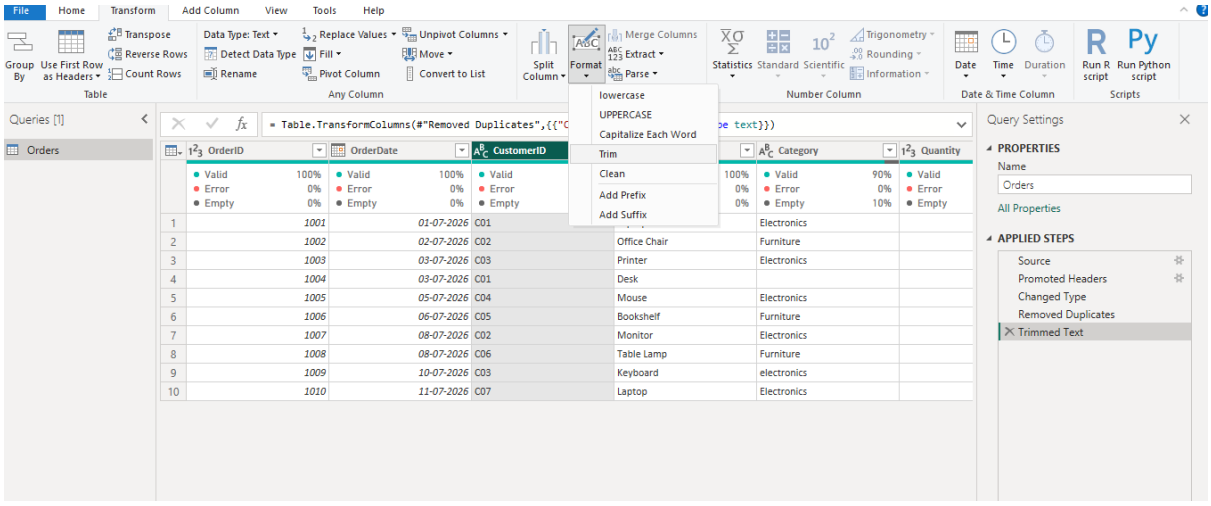
KNIME- https://github.com/Geetanjali-Bhardwaj/Rollno.

Cleaned_Sales- https://github.com/Geetanjali-Bhardwaj/Sales

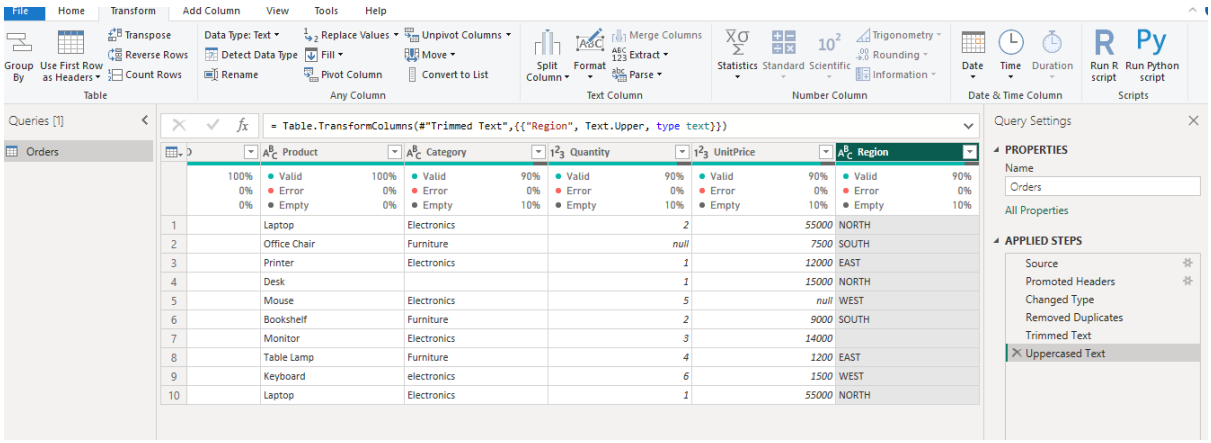
Power BI



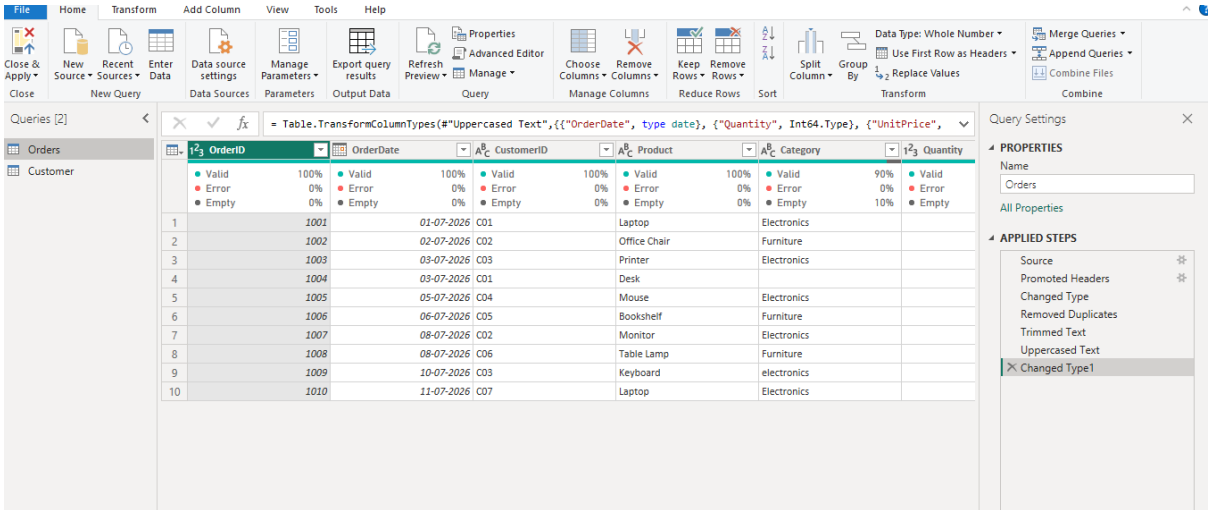




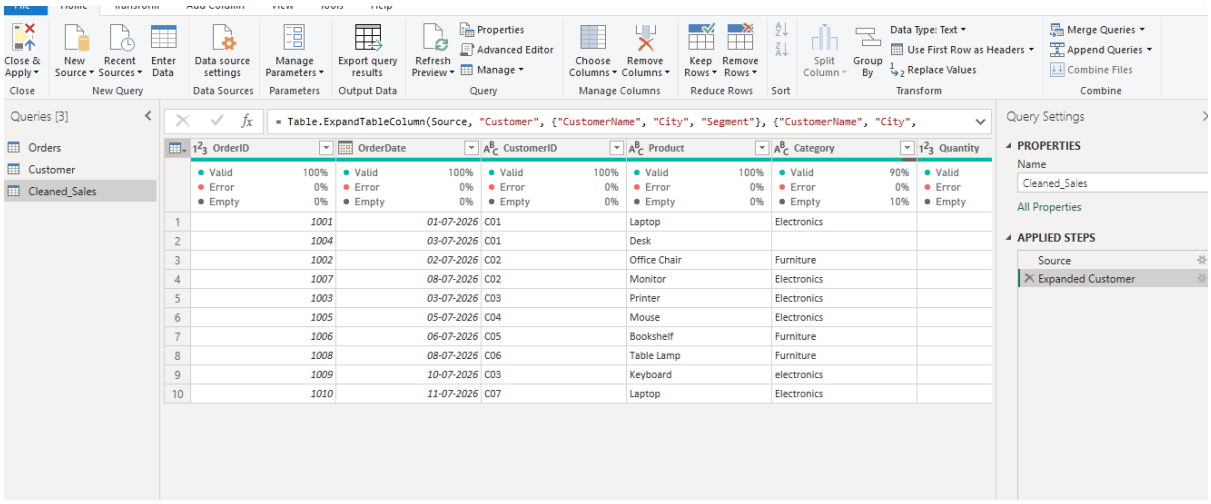
	OrderID	OrderDate	CustomerID		Category	Quantity
1	1001	01-07-2026	C01		Electronics	
2	1002	02-07-2026	C02	Office Chair	Furniture	
3	1003	03-07-2026	C03	Printer	Electronics	
4	1004	03-07-2026	C01	Desk		
5	1005	05-07-2026	C04	Mouse	Electronics	
6	1006	06-07-2026	C05	Bookshelf	Furniture	
7	1007	08-07-2026	C02	Monitor	Electronics	
8	1008	08-07-2026	C06	Table Lamp	Furniture	
9	1009	10-07-2026	C03	Keyboard	electronics	
10	1010	11-07-2026	C07	Laptop	Electronics	



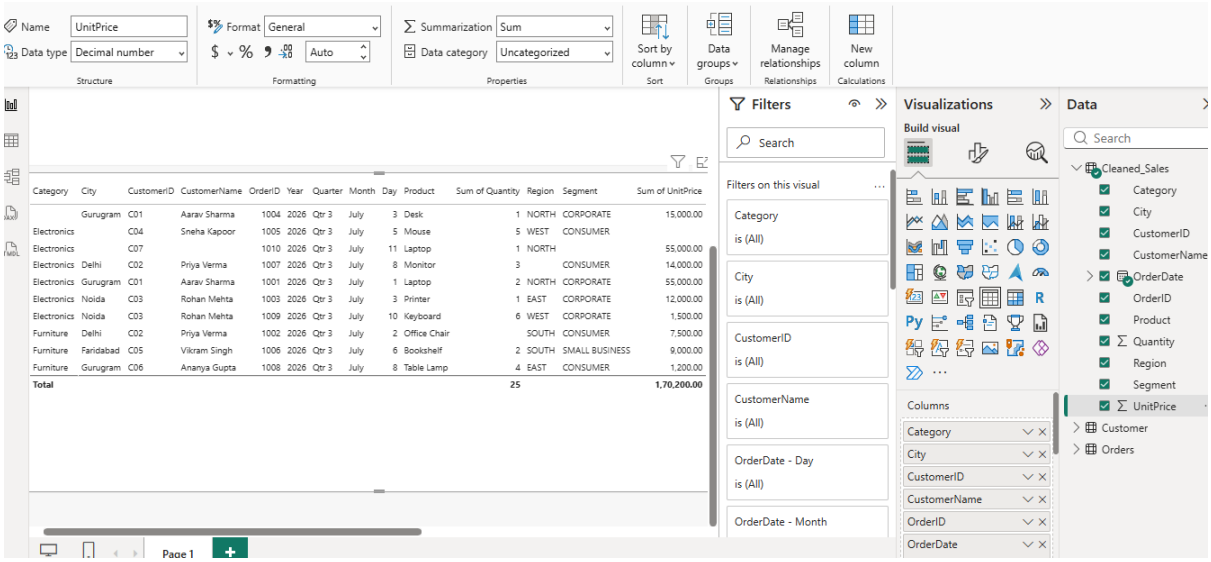
	Product	Category	Quantity	UnitPrice	Region
1	Laptop	Electronics	2	55000	NORTH
2	Office Chair	Furniture	null	7500	SOUTH
3	Printer	Electronics	1	12000	EAST
4	Desk		1	15000	NORTH
5	Mouse	Electronics	5	null	WEST
6	Bookshelf	Furniture	2	9000	SOUTH
7	Monitor	Electronics	3	14000	
8	Table Lamp	Furniture	4	1200	EAST
9	Keyboard	electronics	6	1500	WEST
10	Laptop	Electronics	1	55000	NORTH



	OrderID	OrderDate	CustomerID	Product	Category	Quantity
1	1001	01-07-2026	C01	Laptop	Electronics	
2	1002	02-07-2026	C02	Office Chair	Furniture	
3	1003	03-07-2026	C03	Printer	Electronics	
4	1004	03-07-2026	C01	Desk		
5	1005	05-07-2026	C04	Mouse	Electronics	
6	1006	06-07-2026	C05	Bookshelf	Furniture	
7	1007	08-07-2026	C02	Monitor	Electronics	
8	1008	08-07-2026	C06	Table Lamp	Furniture	
9	1009	10-07-2026	C03	Keyboard	electronics	
10	1010	11-07-2026	C07	Laptop	Electronics	



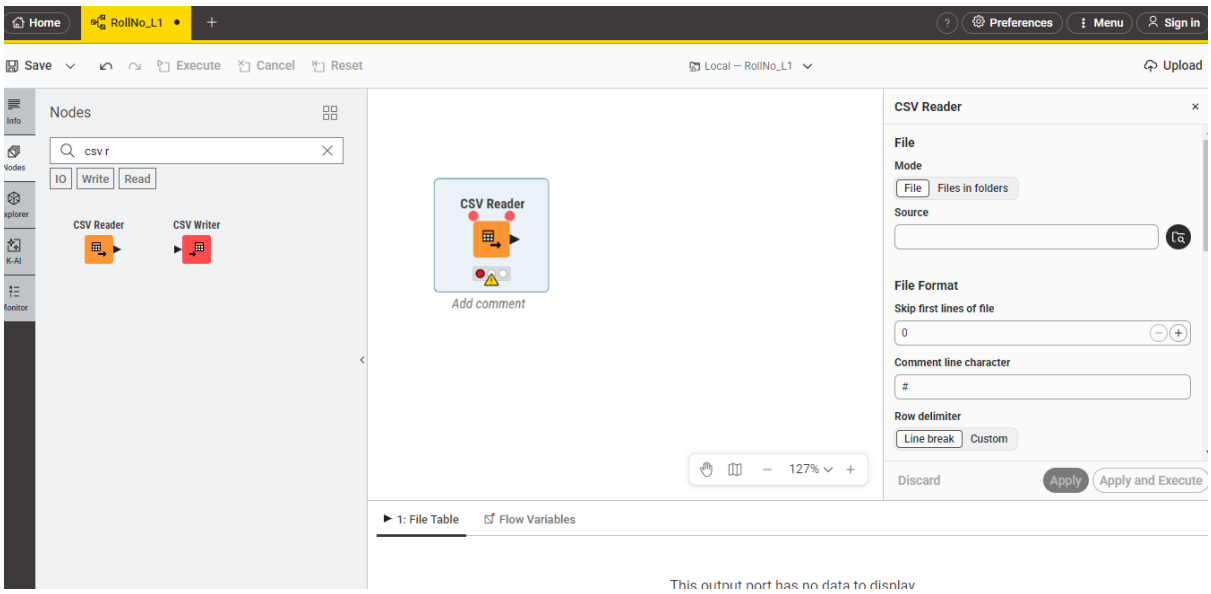
The Final Screen:-



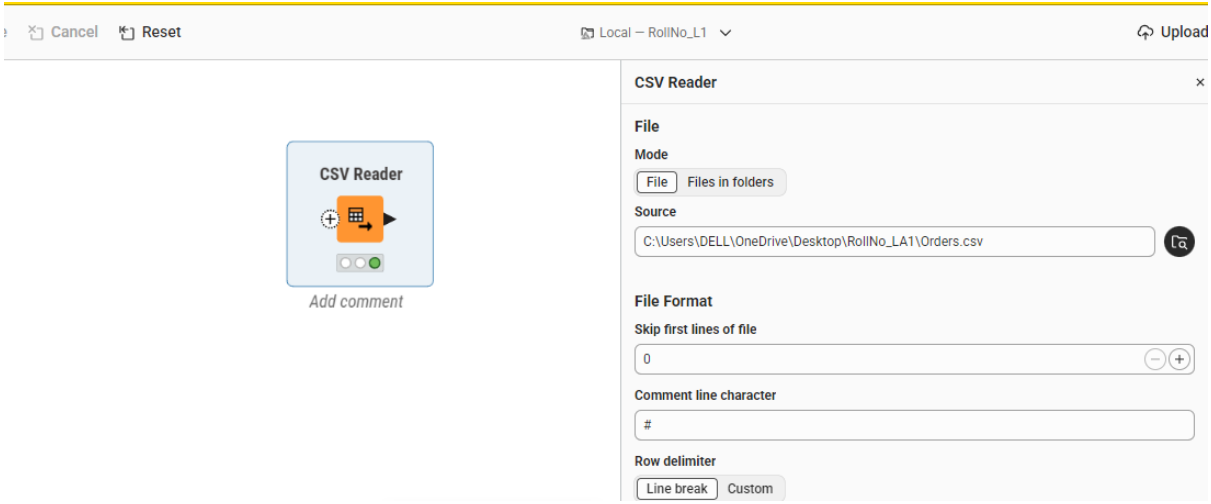
Category	City	CustomerID	CustomerName	OrderID	Year	Quarter	Month	Day	Product	Sum of Quantity	Region	Segment	Sum of UnitPrice
	Gurugram	C01	Aarav Sharma	1004	2026	Qtr 3	July	3	Desk	1	NORTH	CORPORATE	15,000.00
Electronics		C04	Sneha Kapoor	1005	2026	Qtr 3	July	5	Mouse	5	WEST	CONSUMER	
Electronics		C07		1010	2026	Qtr 3	July	11	Laptop	1	NORTH		55,000.00
Electronics	Delhi	C02	Priya Verma	1007	2026	Qtr 3	July	8	Monitor	3		CONSUMER	14,000.00
Electronics	Gurugram	C01	Aarav Sharma	1001	2026	Qtr 3	July	1	Laptop	2	NORTH	CORPORATE	55,000.00
Electronics	Noida	C03	Rohan Mehta	1003	2026	Qtr 3	July	3	Printer	1	EAST	CORPORATE	12,000.00
Electronics	Noida	C03	Rohan Mehta	1009	2026	Qtr 3	July	10	Keyboard	6	WEST	CORPORATE	1,500.00
Furniture	Delhi	C02	Priya Verma	1002	2026	Qtr 3	July	2	Office Chair		SOUTH	CONSUMER	7,500.00
Furniture	Faridabad	C05	Vikram Singh	1006	2026	Qtr 3	July	6	Bookshelf	2	SOUTH	SMALL BUSINESS	9,000.00
Furniture	Gurugram	C06	Ananya Gupta	1008	2026	Qtr 3	July	8	Table Lamp	4	EAST	CONSUMER	1,200.00
Total										**25**			**1,70,200.00**

KNIME

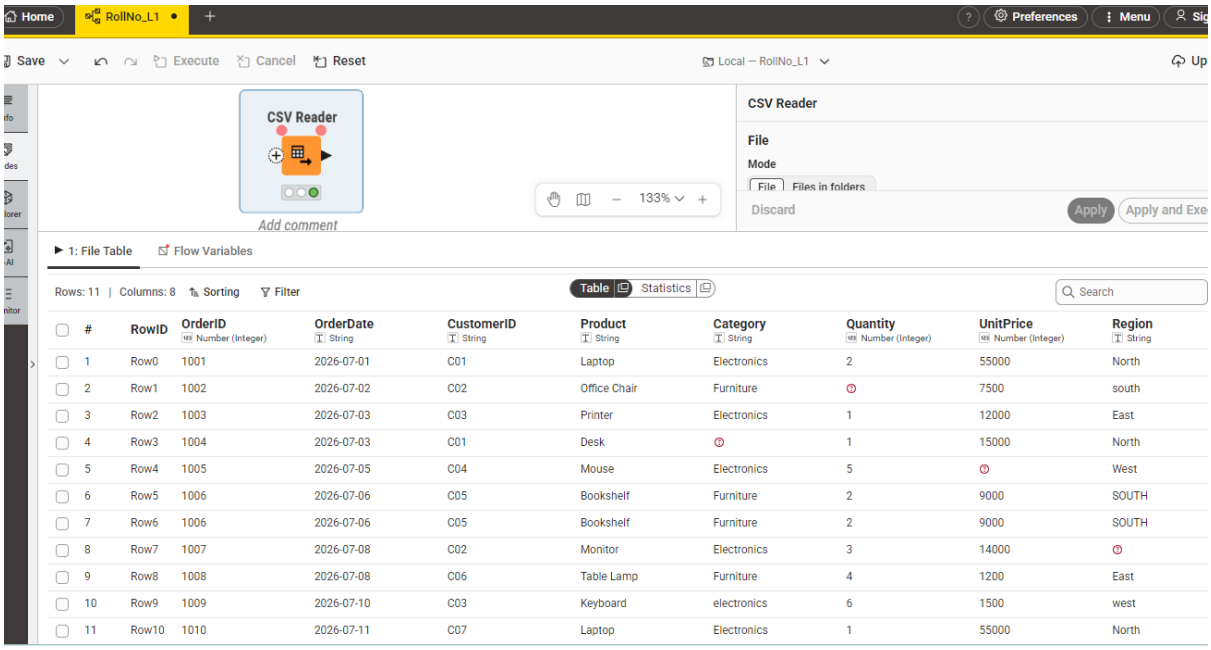
Step 1. Search and drag CSV Reader Node from the nodes.



Step 2. Add Configurations to CSV Reader node, by browsing the file named ‘Orders.csv’

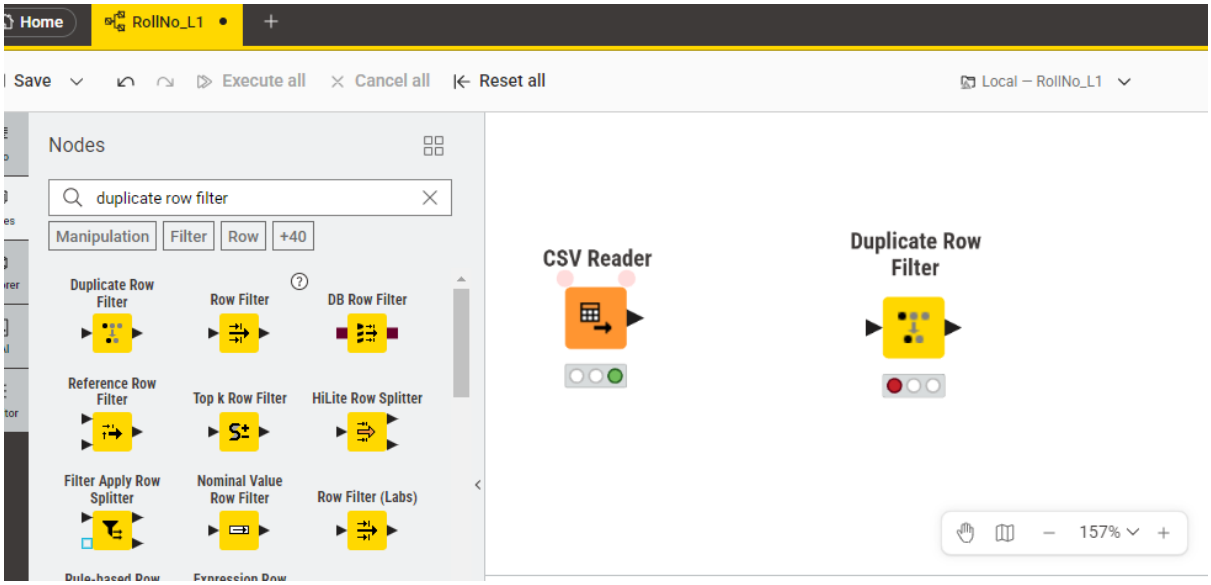


Step 3. Apply and Execute.

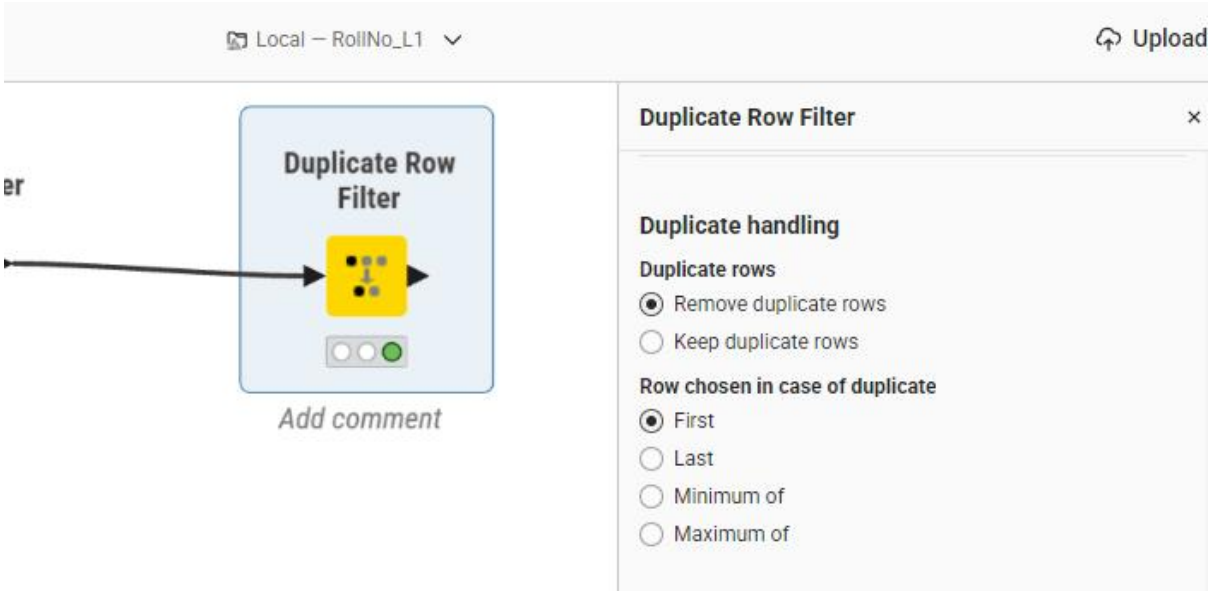


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	?	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row3	1004	2026-07-03	C01	Desk	?	1	15000	North
5	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
6	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
7	Row6	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
8	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
9	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
10	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
11	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

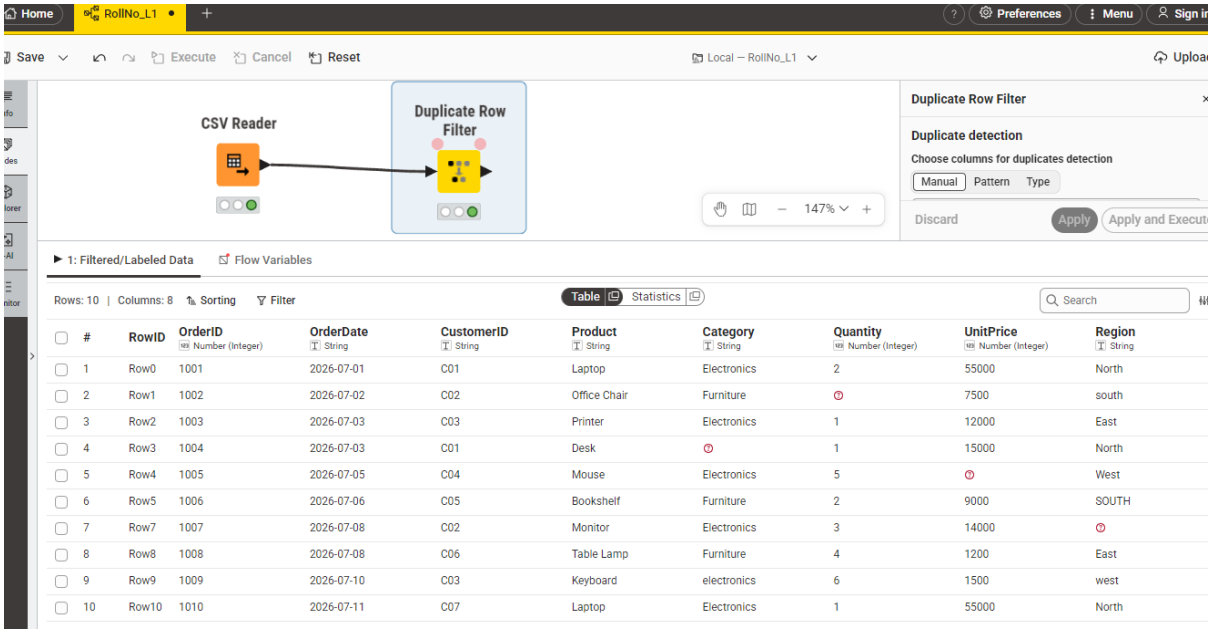
Step 4. Drag Duplicate Row Filter node , to remove duplicate rows from the orders.csv.



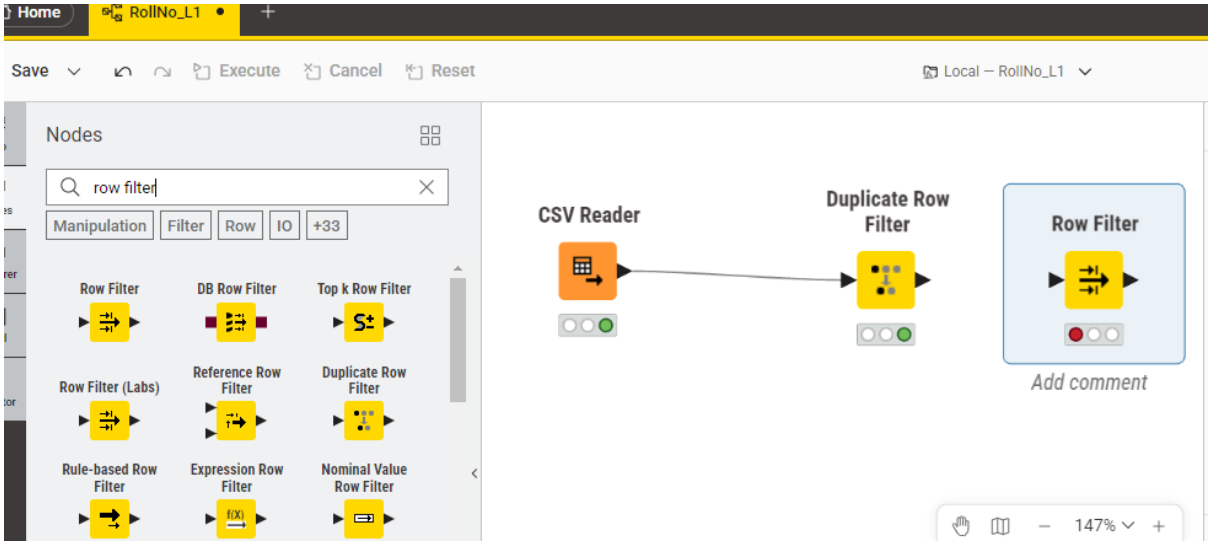
Step 5. Join it to CSV Reader node , then add configurations to this node.

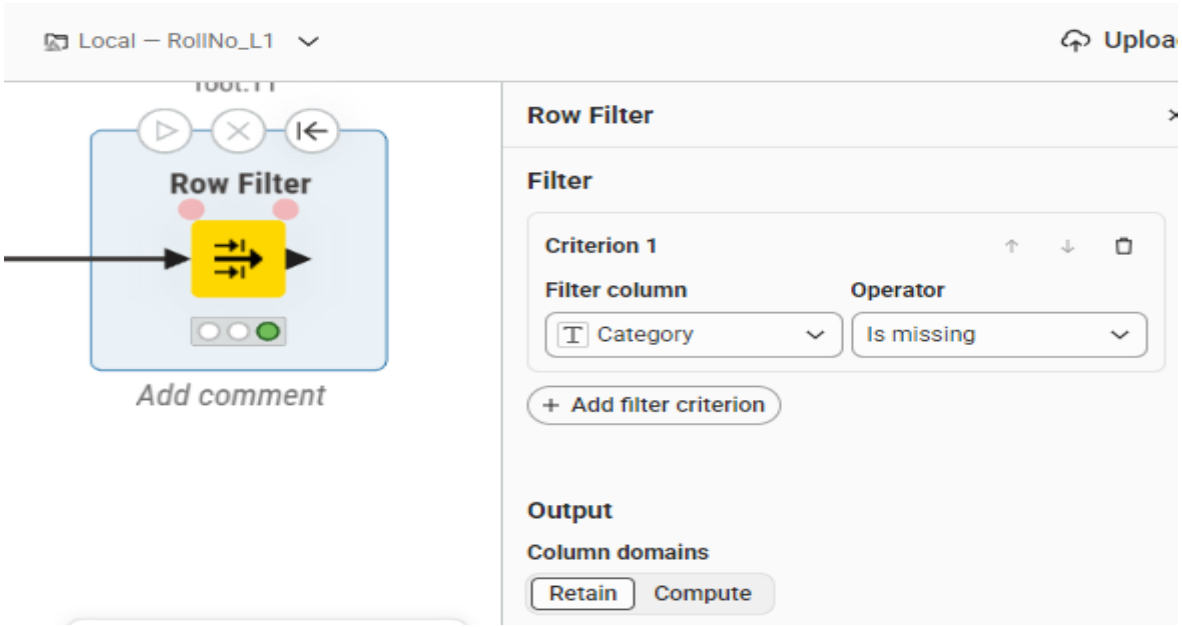


Step 6. Execute the node.



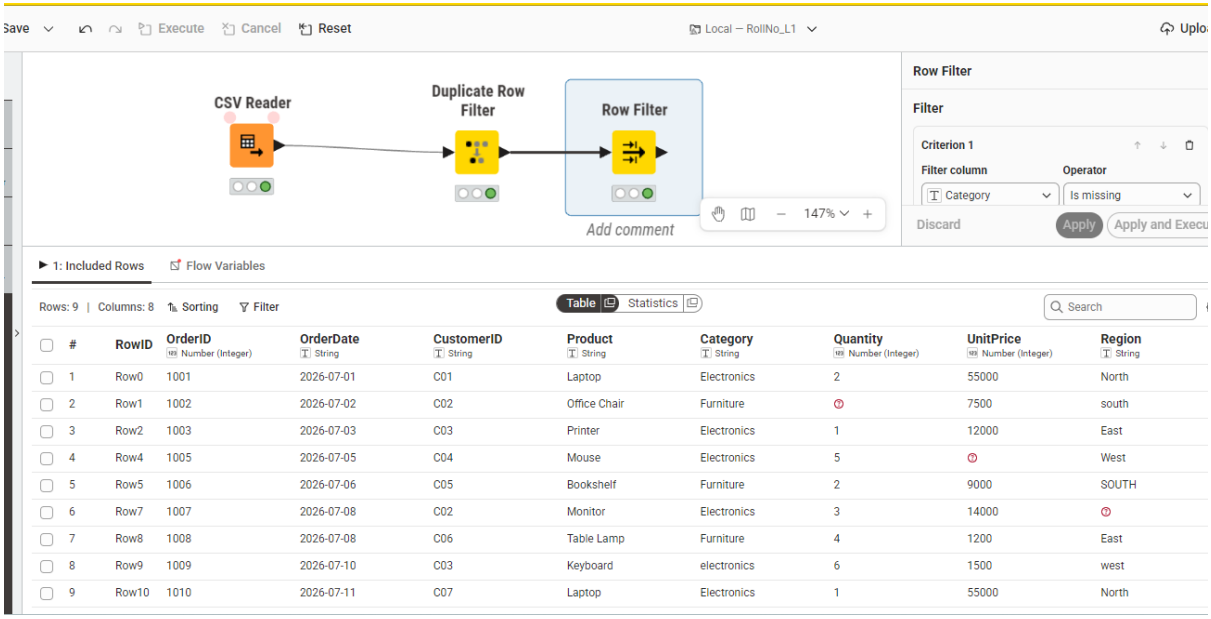
#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region
		Number (Integer)	String	String	String	String	Number (Integer)	Number (Integer)	String
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	?	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row3	1004	2026-07-03	C01	Desk	?	1	15000	North
5	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
6	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
7	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
8	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
9	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
10	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North





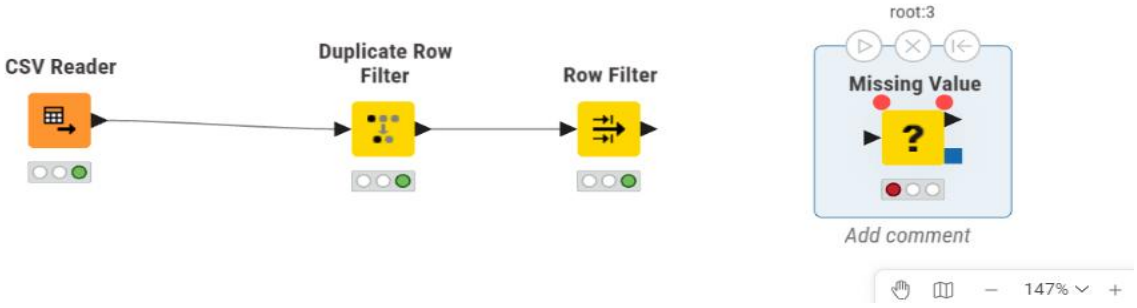
Step 7. Drag Row Filter node and connect it to Duplicate Row Filter.

Add Configuration and execute it.

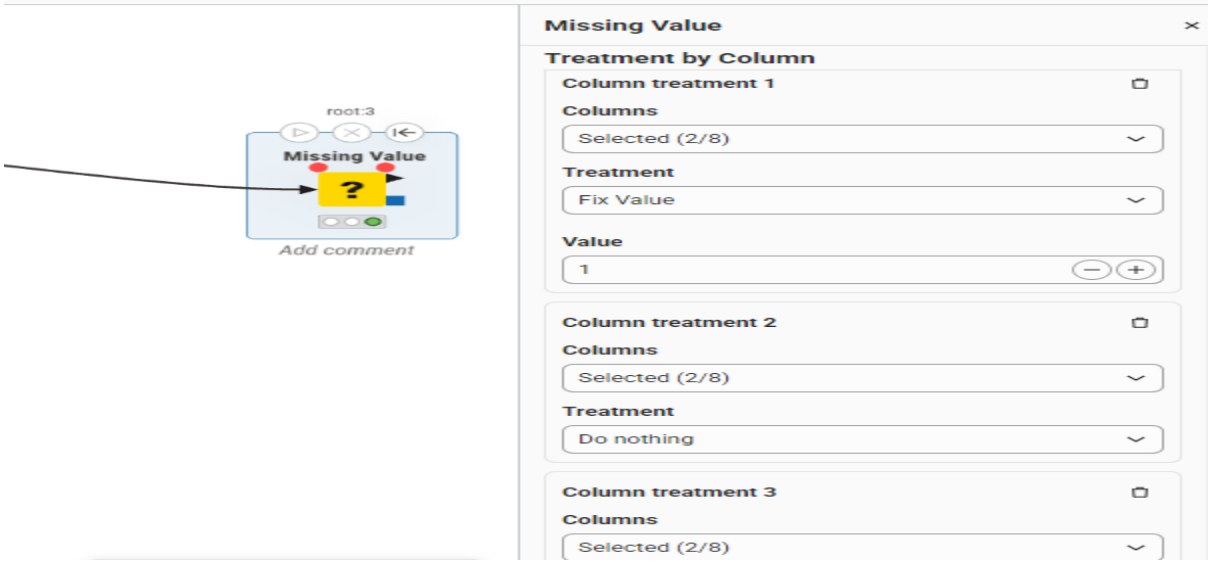


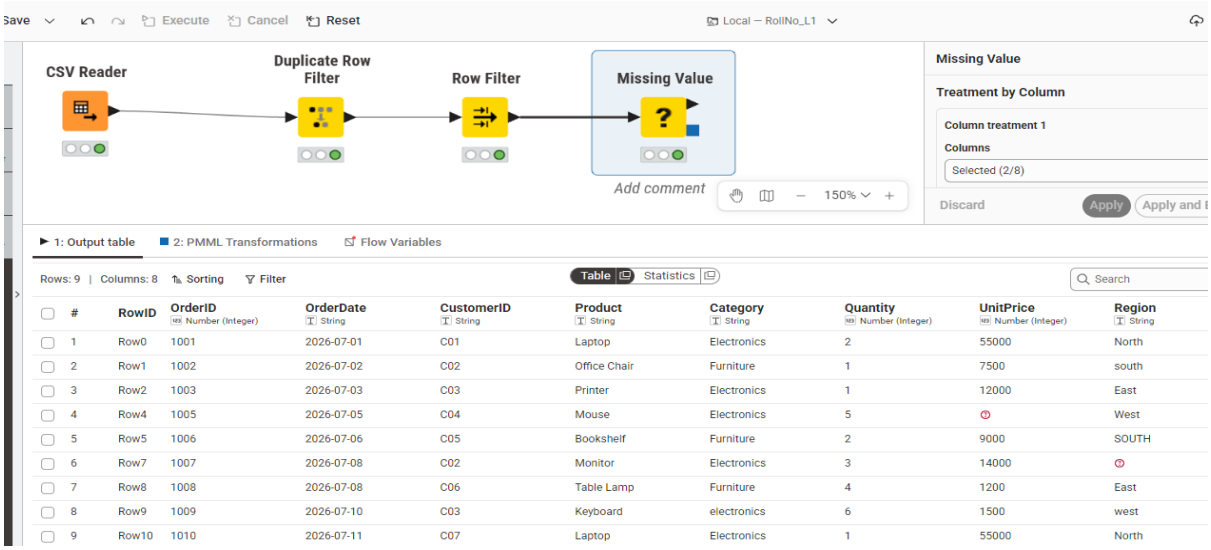
#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	?	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

Step 8. Drag Missing Value.



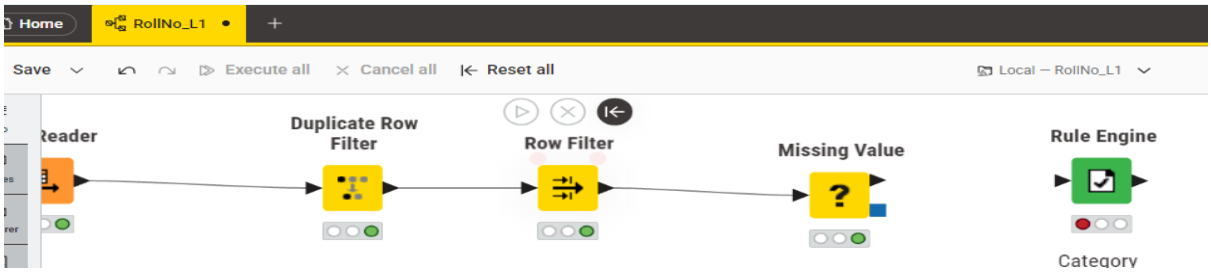
Step 9. Add Column Treatment 3 times for CustomerID, Category and Quantity. Execute.



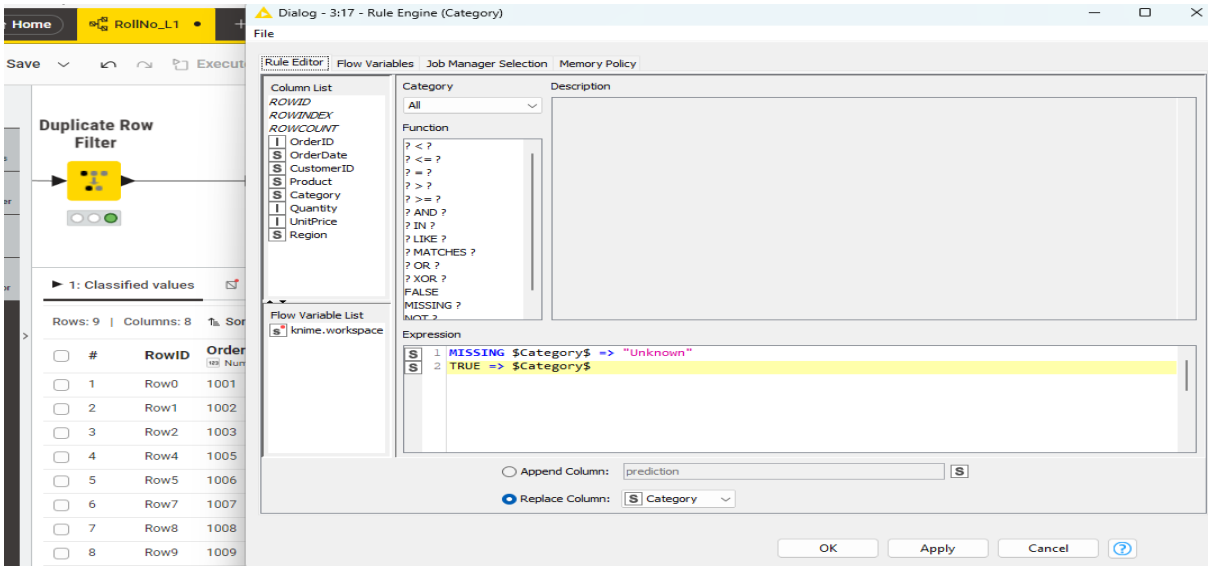


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

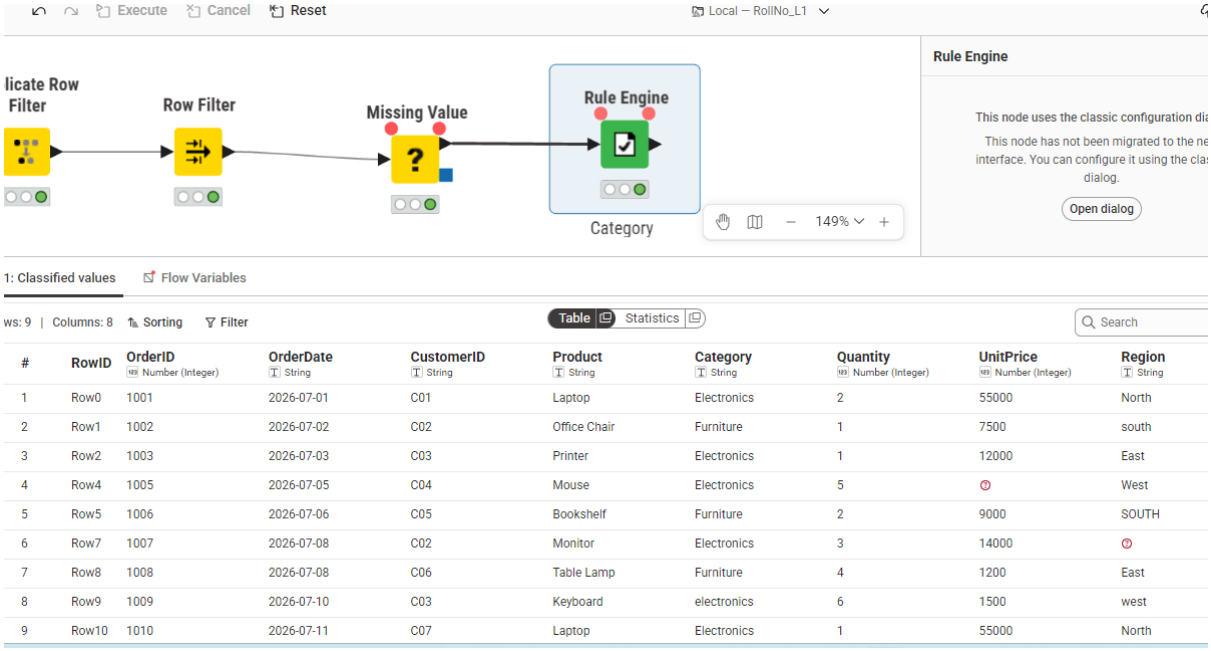
Step 10. Search and drag Rule Engine node. Connect it to Missing Value.



Step 11. Double click on Rule Engine and write under expression for Category.

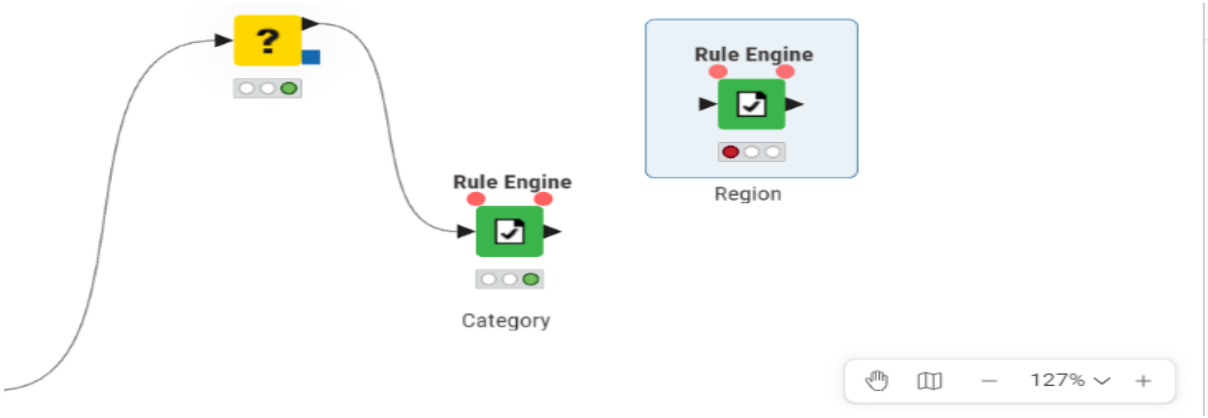


Step 12. Execute.

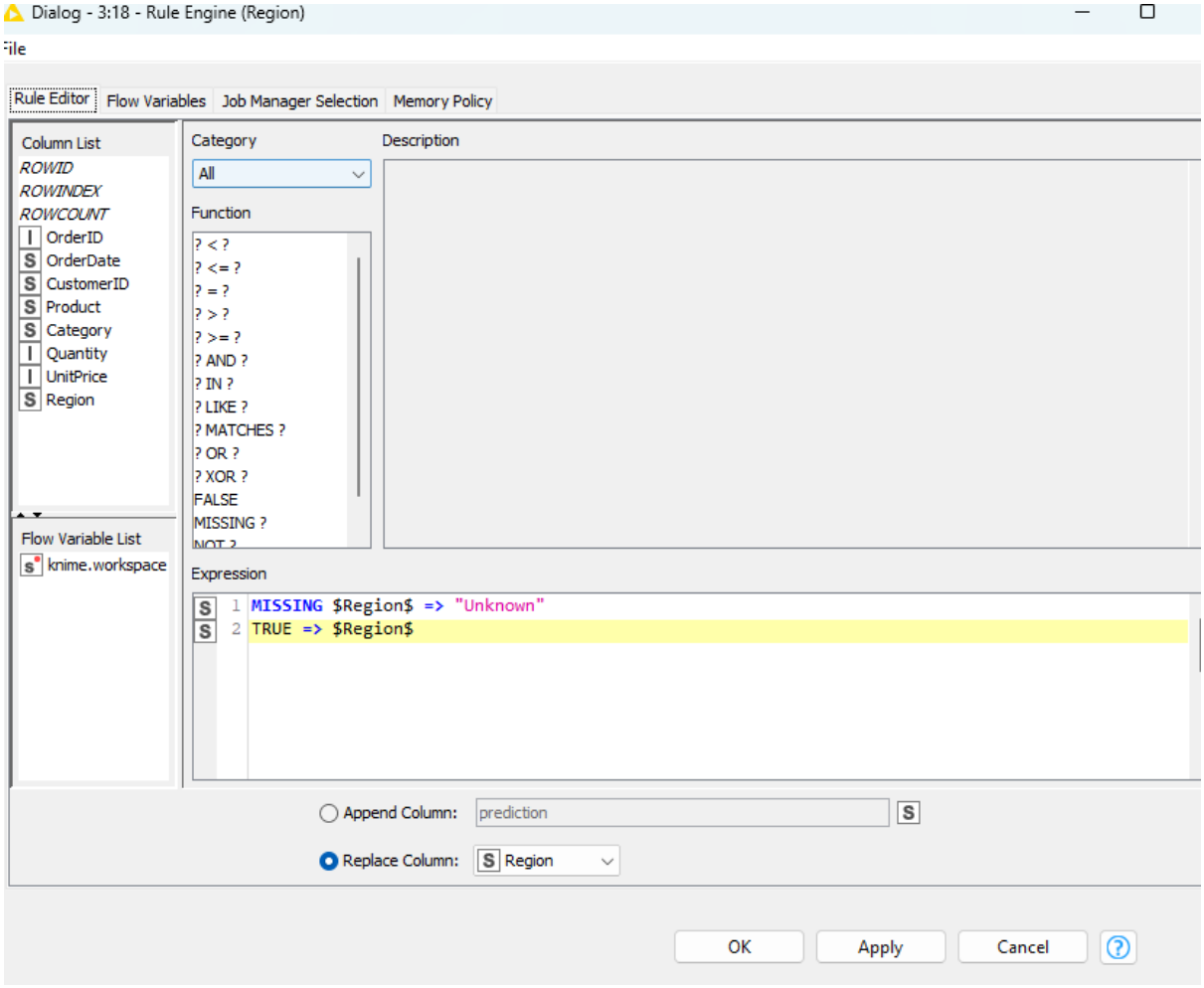


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5	?	West
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	?
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

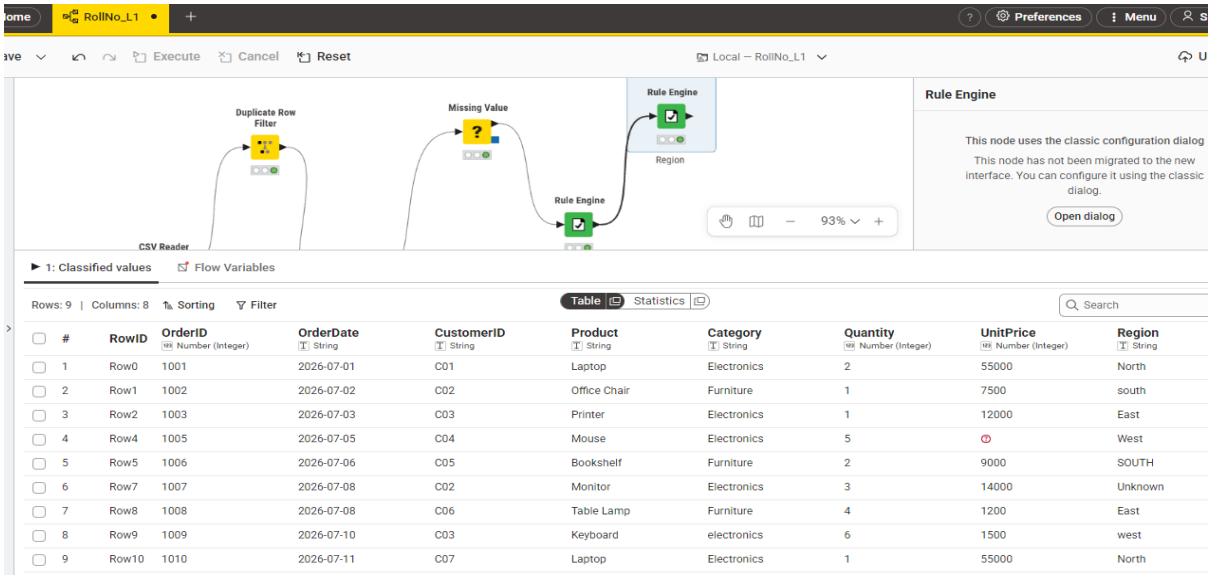
Step 13. Drag another Rule Engine.



Step 14. Double click on Rule Engine and write under expression for Region.

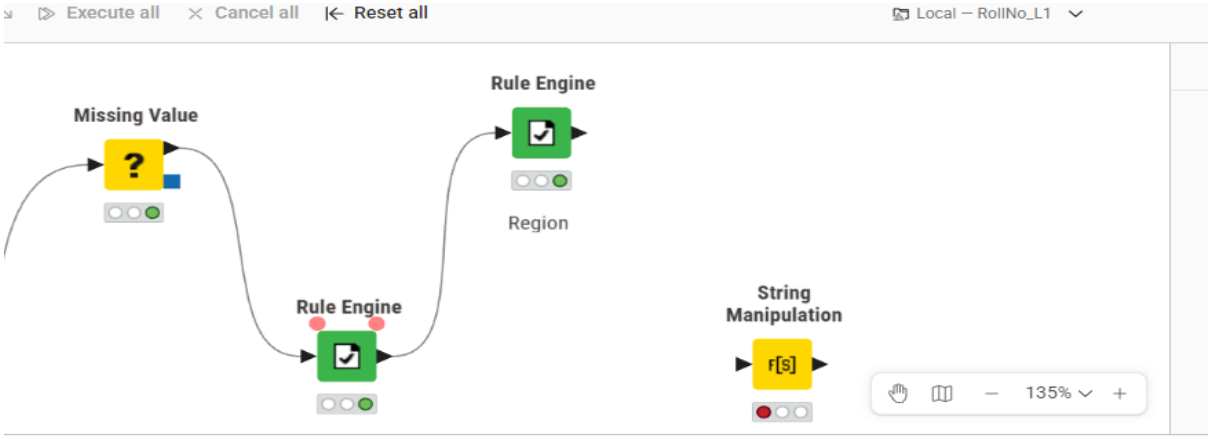


Step 15. Execute

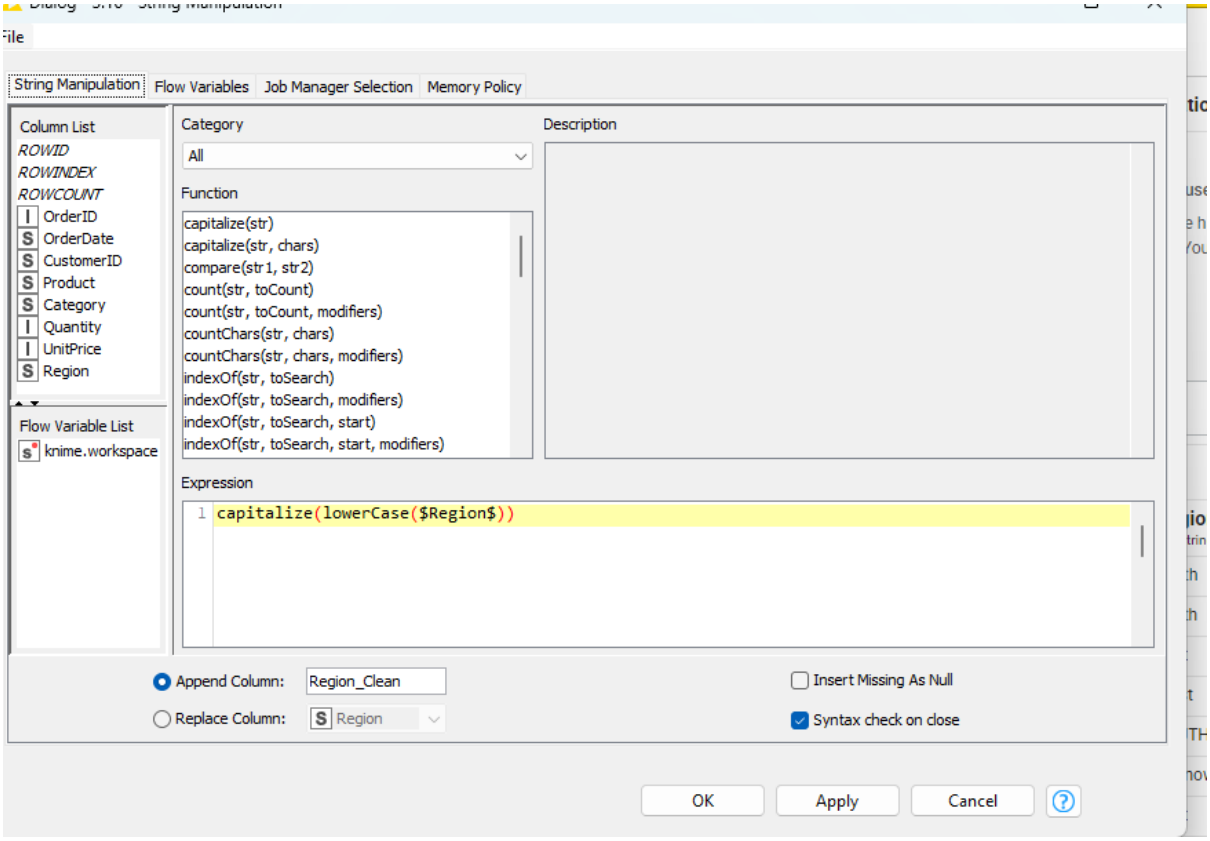


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5		West
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North

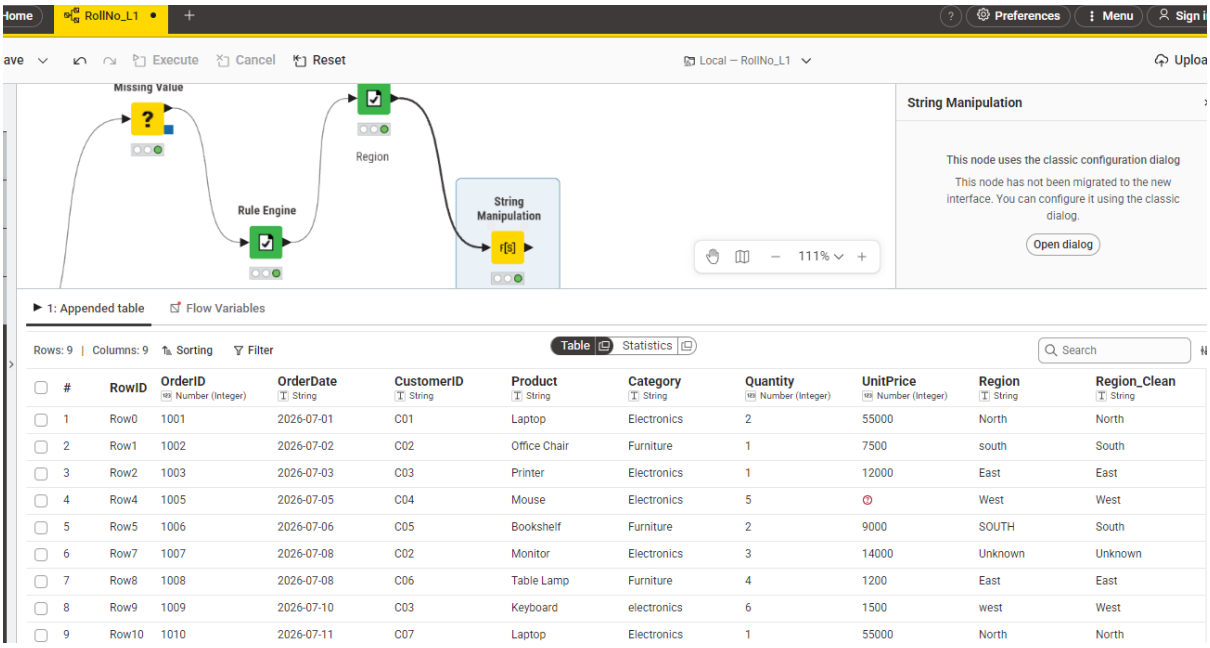
Step 16. Drag String Manipulation node. Connect it to Rule Engine.



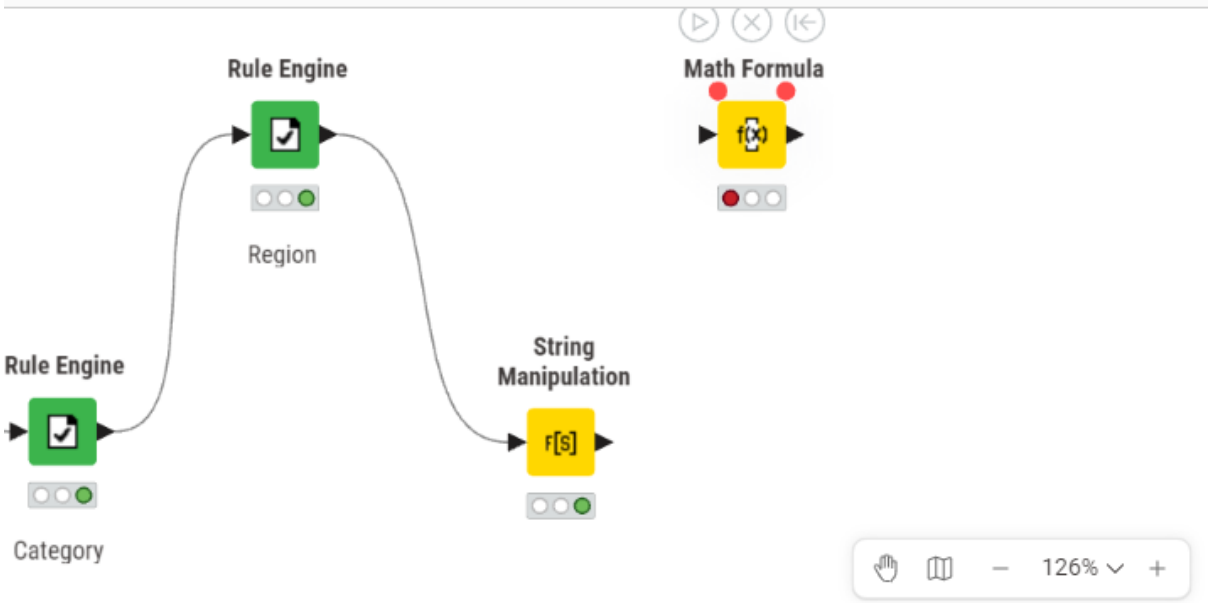
Step 17. Double click on the node and write:

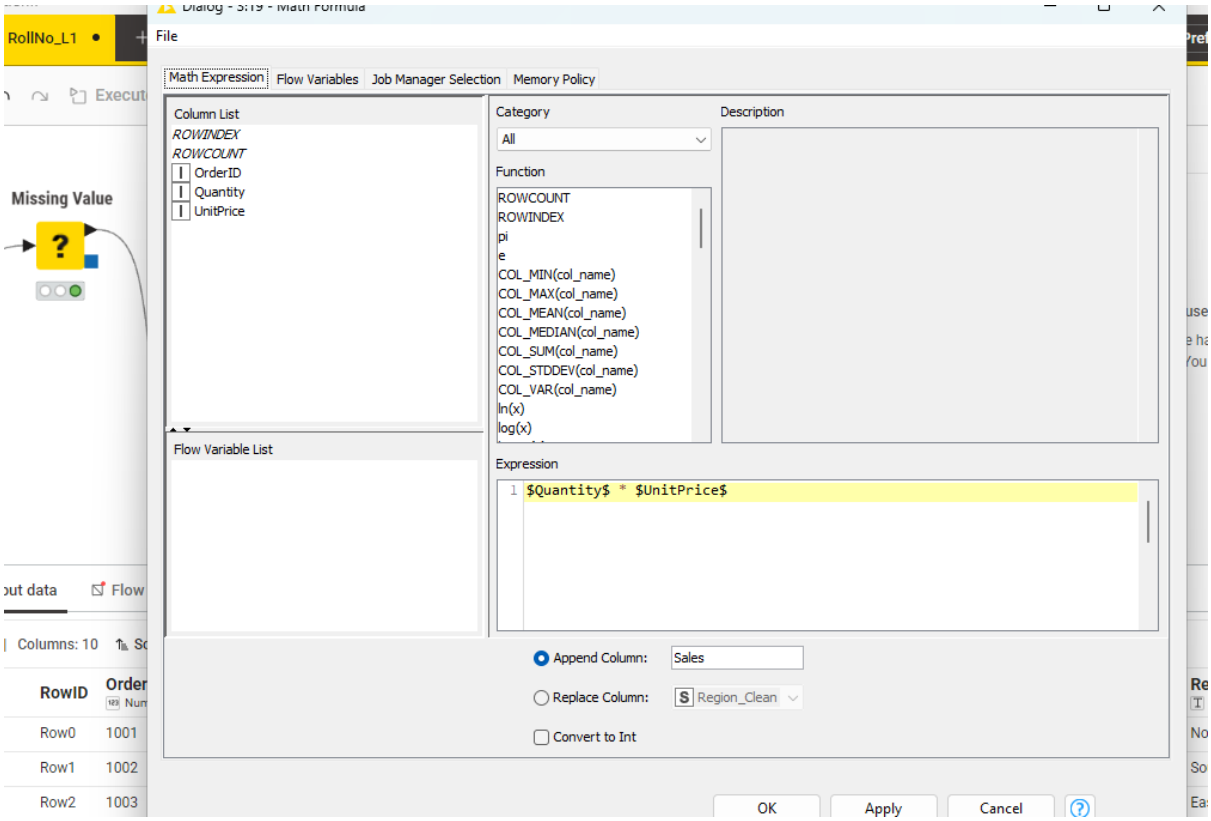


Step 18. Execute and check the region become standardized lower case.

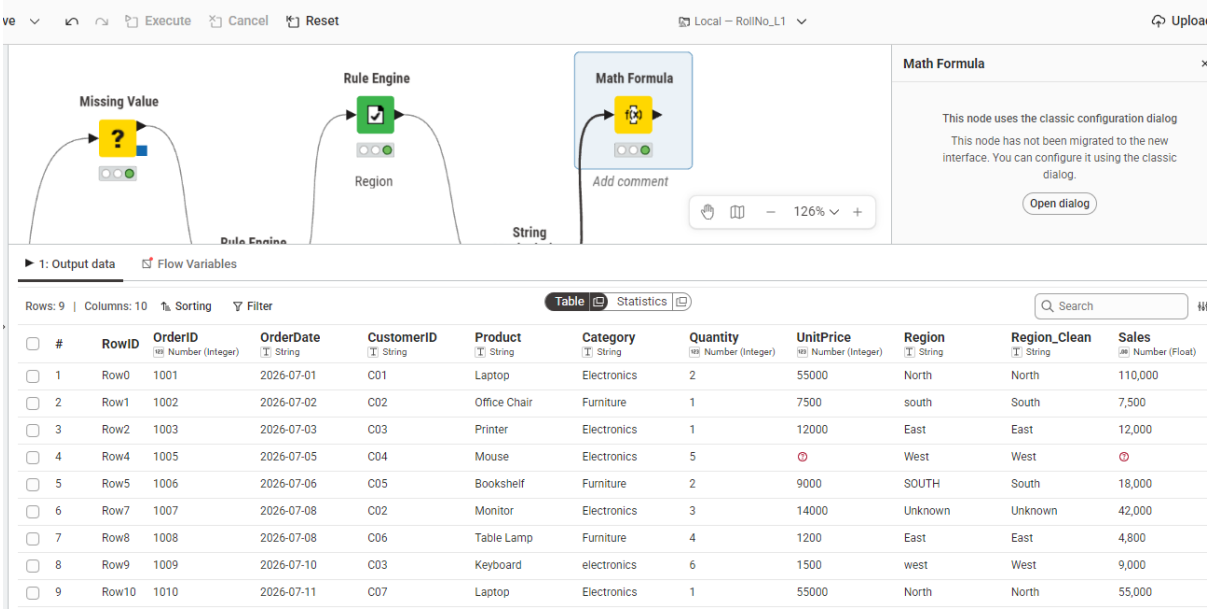


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Region_Clean
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5		West	West
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North	North


Rule Engine
Region
Math Formula
Rule Engine
Category
String Manipulation
126%

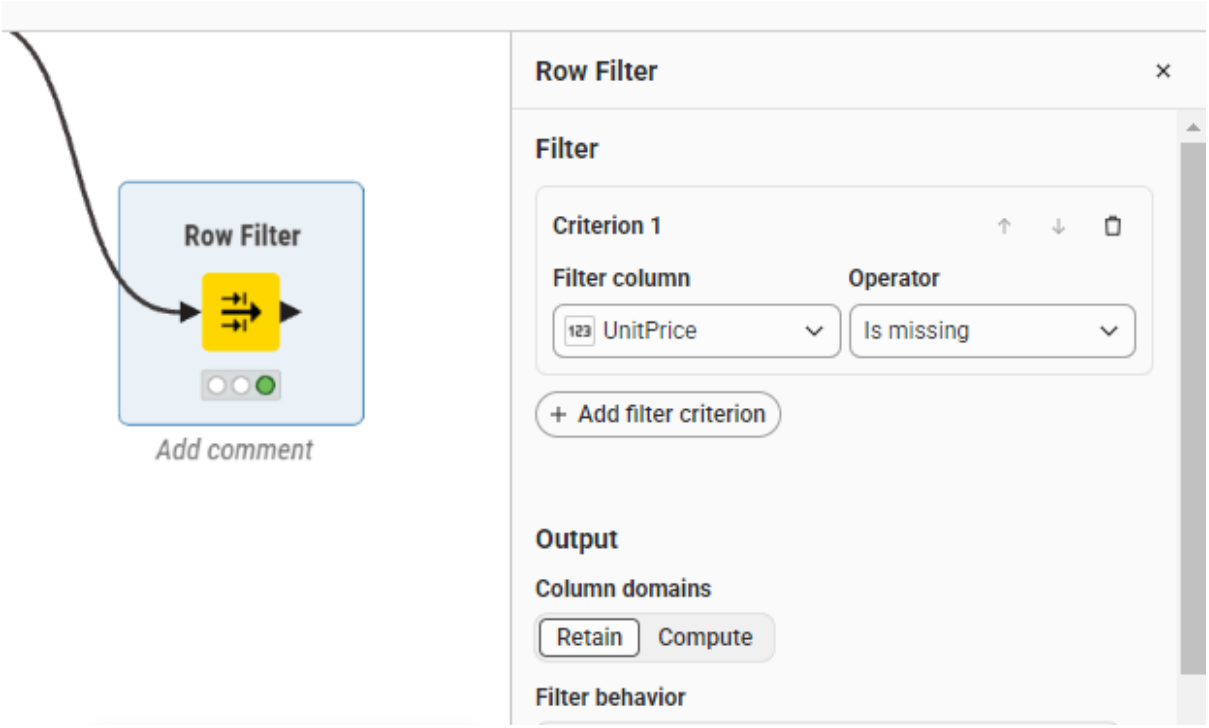

RollNo_L1
File
Math Expression
Flow Variables
Job Manager Selection
Memory Policy
Column List
ROWINDEX
ROWCOUNT
OrderID
Quantity
UnitPrice
Category
All
Description
Function
ROWCOUNT
ROWINDEX
pi
e
COL_MIN(col_name)
COL_MAX(col_name)
COL_MEAN(col_name)
COL_MEDIAN(col_name)
COL_SUM(col_name)
COL_STDDEV(col_name)
COL_VAR(col_name)
ln(x)
log(x)
Flow Variable List
Expression
$Quantity$ * $UnitPrice$
Missing Value
Append Column:
Sales
Replace Column:
Region_Clean
Convert to Int
OK
Apply
Cancel
Columns: 10
RowID
Row0
Row1
Row2
1001
1002
1003

Step 19. Drag Math Formula node connect it to String Manipulation.

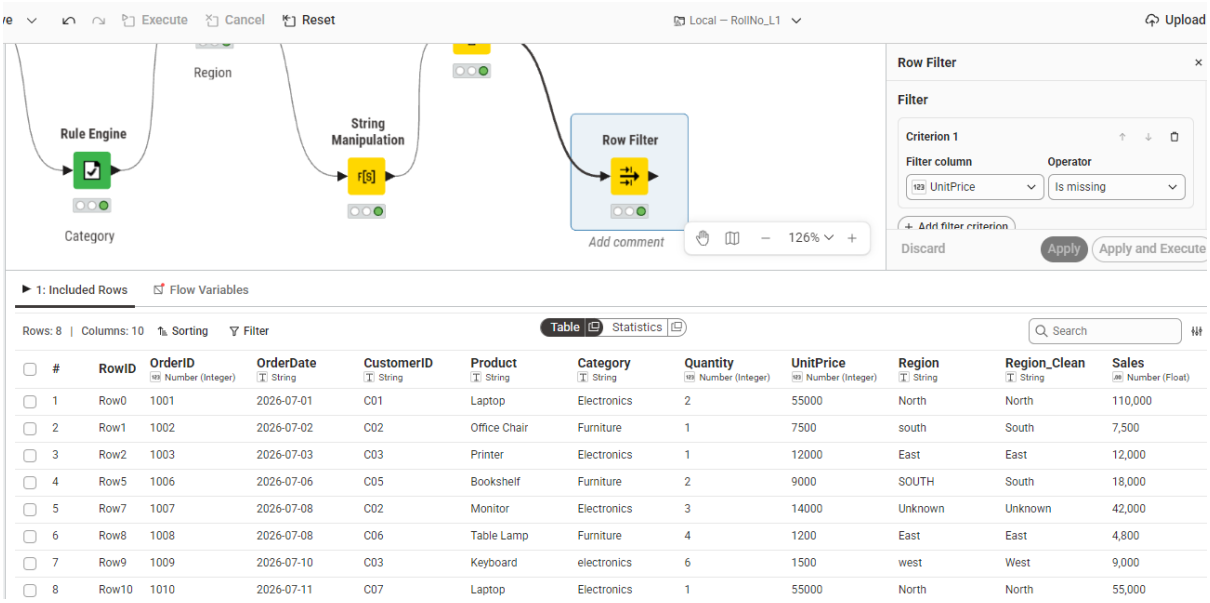


#	RowID	OrderID (Number (Integer))	OrderDate (String)	CustomerID (String)	Product (String)	Category (String)	Quantity (Number (Integer))	UnitPrice (Number (Integer))	Region (String)	Region_Clean (String)	Sales (Number (Float))
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North	110,000
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South	7,500
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East	12,000
4	Row4	1005	2026-07-05	C04	Mouse	Electronics	5		West	West	
5	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South	18,000
6	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown	42,000
7	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East	4,800
8	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West	9,000
9	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North	North	55,000

Step 20. Drag Row Filter.

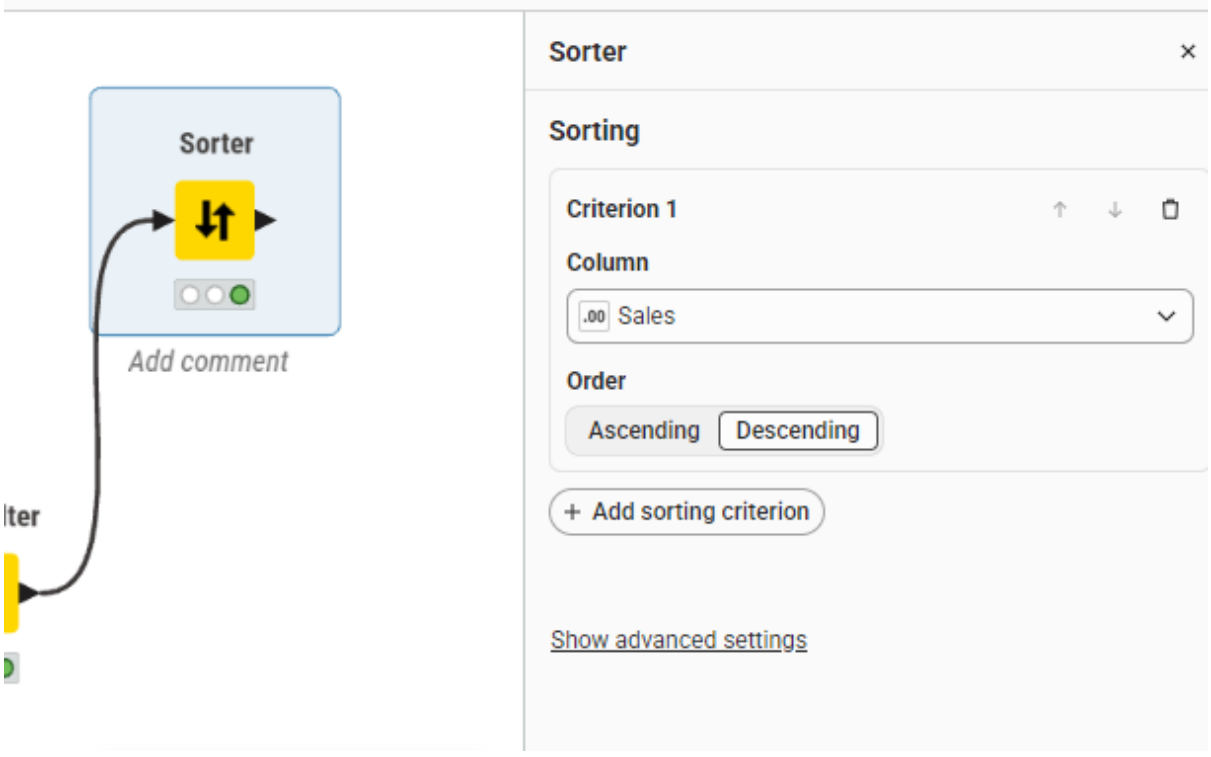


Step 21. Execute Row Filter.

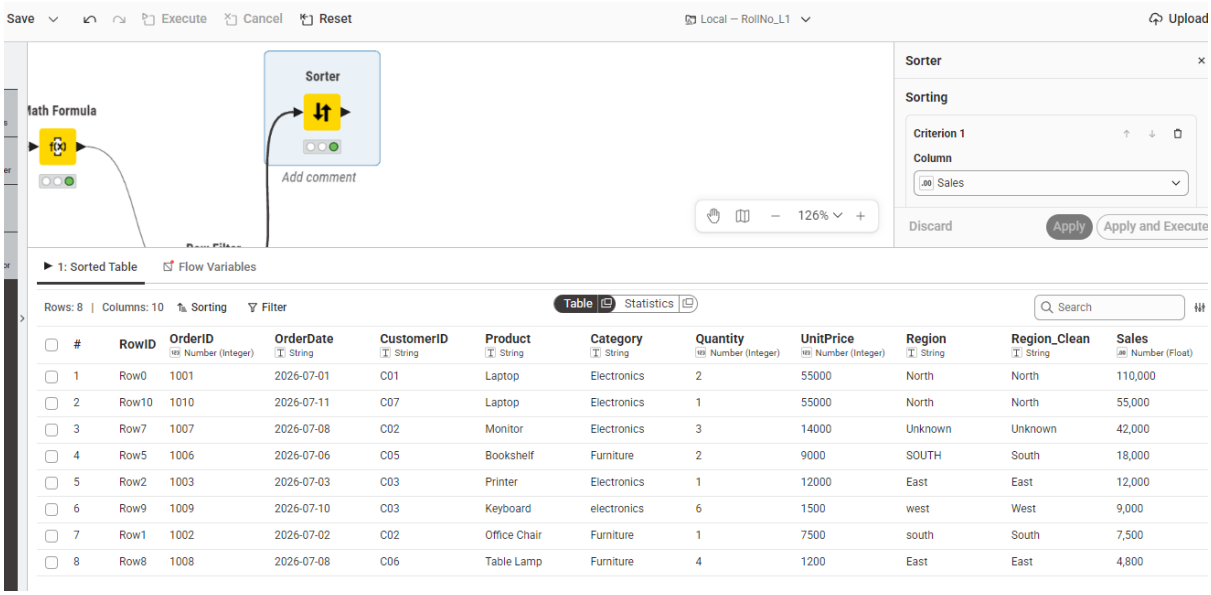


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Region_Clean	Sales
		Number (Integer)	String	String	String	String	Number (Integer)	Number (Integer)	String	String	Number (Float)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North	110,000
2	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South	7,500
3	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East	12,000
4	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South	18,000
5	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown	42,000
6	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East	4,800
7	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West	9,000
8	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North	North	55,000

Step 22. Drag Sorter node. It will sort sales column and show it in descending order.

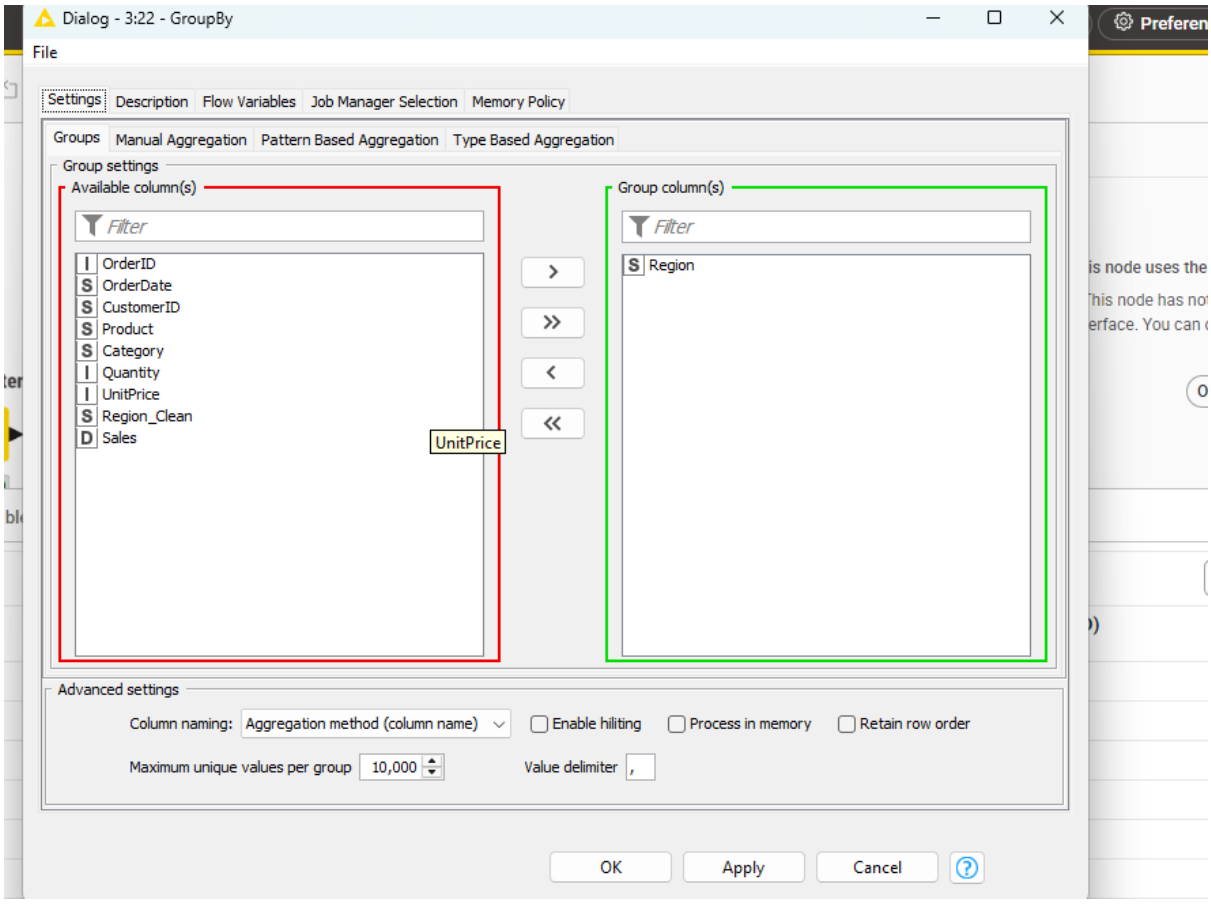


Step 23. We can see the order of sales.

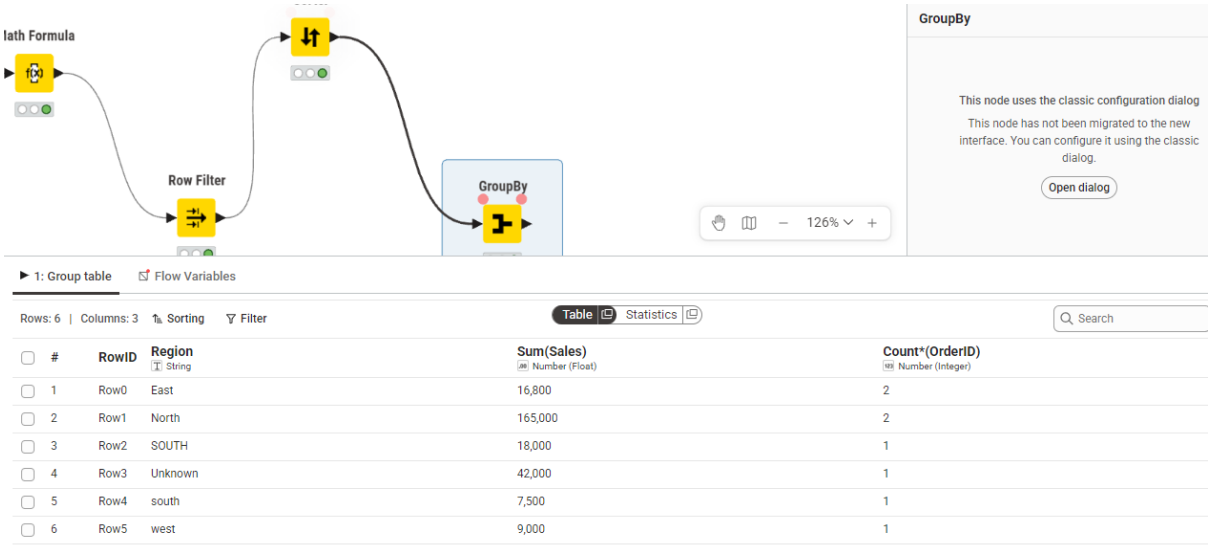


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Region_Clean	Sales
		Number (Integer)	String	String	String	String	Number (Integer)	Number (Integer)	String	String	Number (Float)
1	Row0	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North	110,000
2	Row10	1010	2026-07-11	C07	Laptop	Electronics	1	55000	North	North	55,000
3	Row7	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown	42,000
4	Row5	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South	18,000
5	Row2	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East	12,000
6	Row9	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West	9,000
7	Row1	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South	7,500
8	Row8	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East	4,800

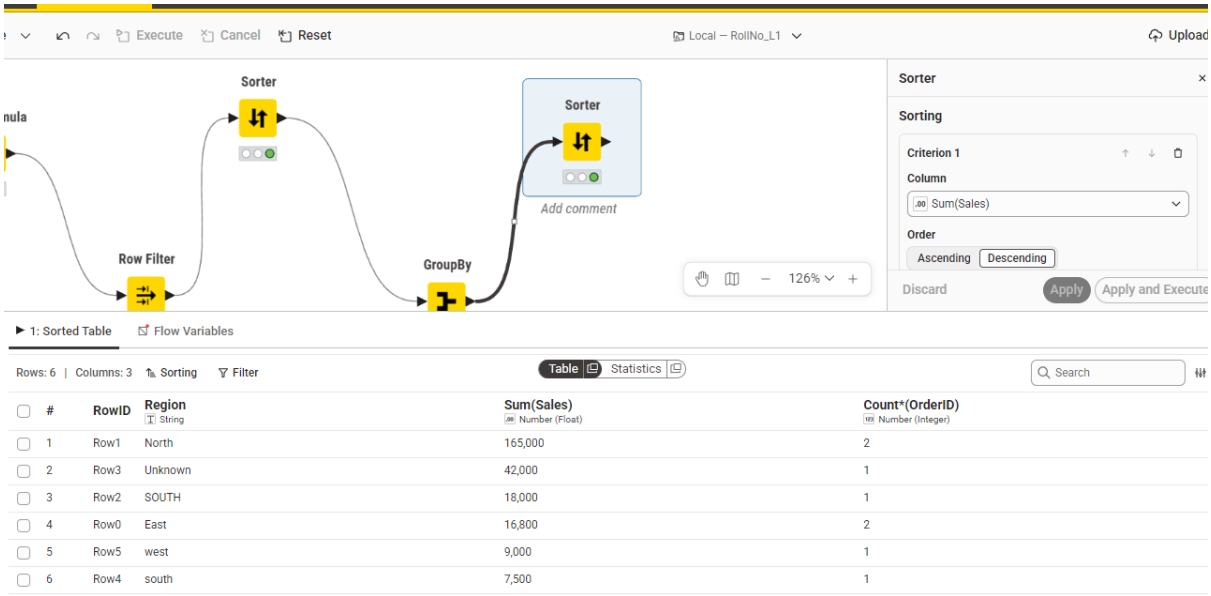
Step 24. In GroupBy node, add configurations.



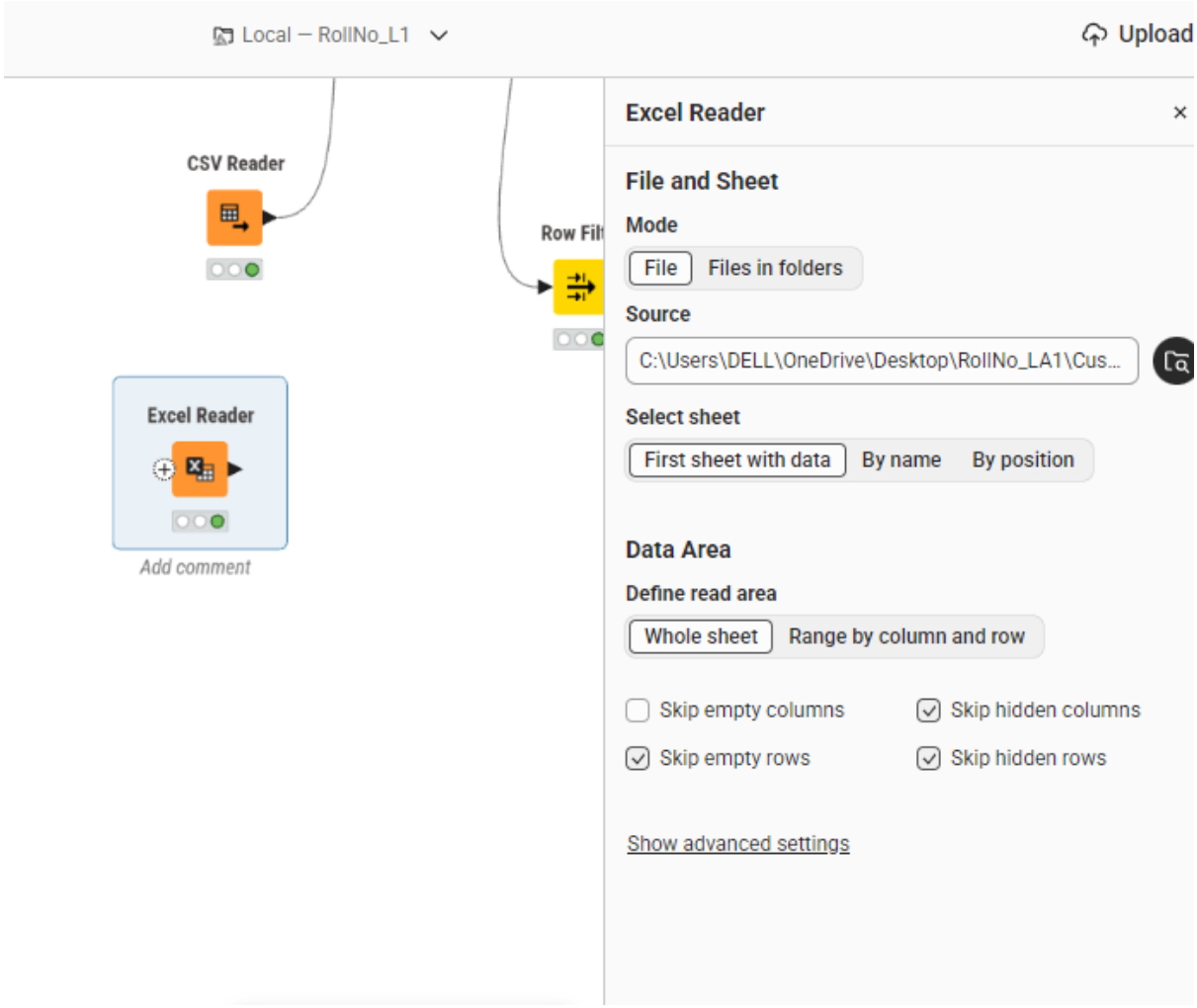
Step 25. Execute.



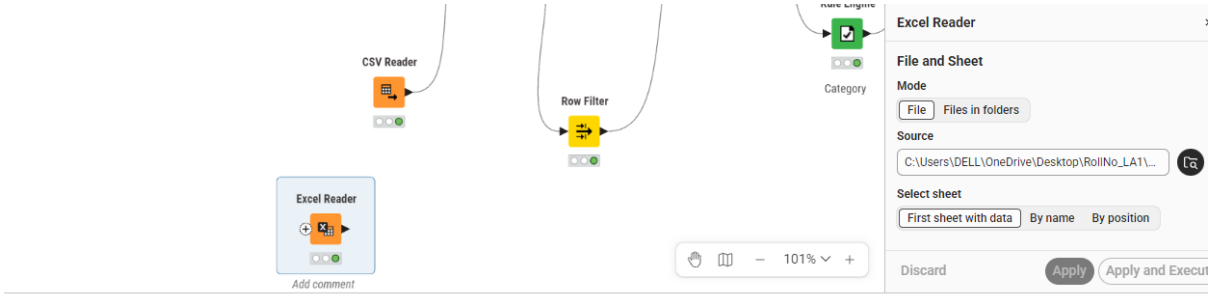
Step 26. Add another Sorter. Connect it to GroupBy, sort Sales and execute.



Step 27. Search and Drag Excel Reader node. Browse Customer.xlsx file.



Step 28. Execute.

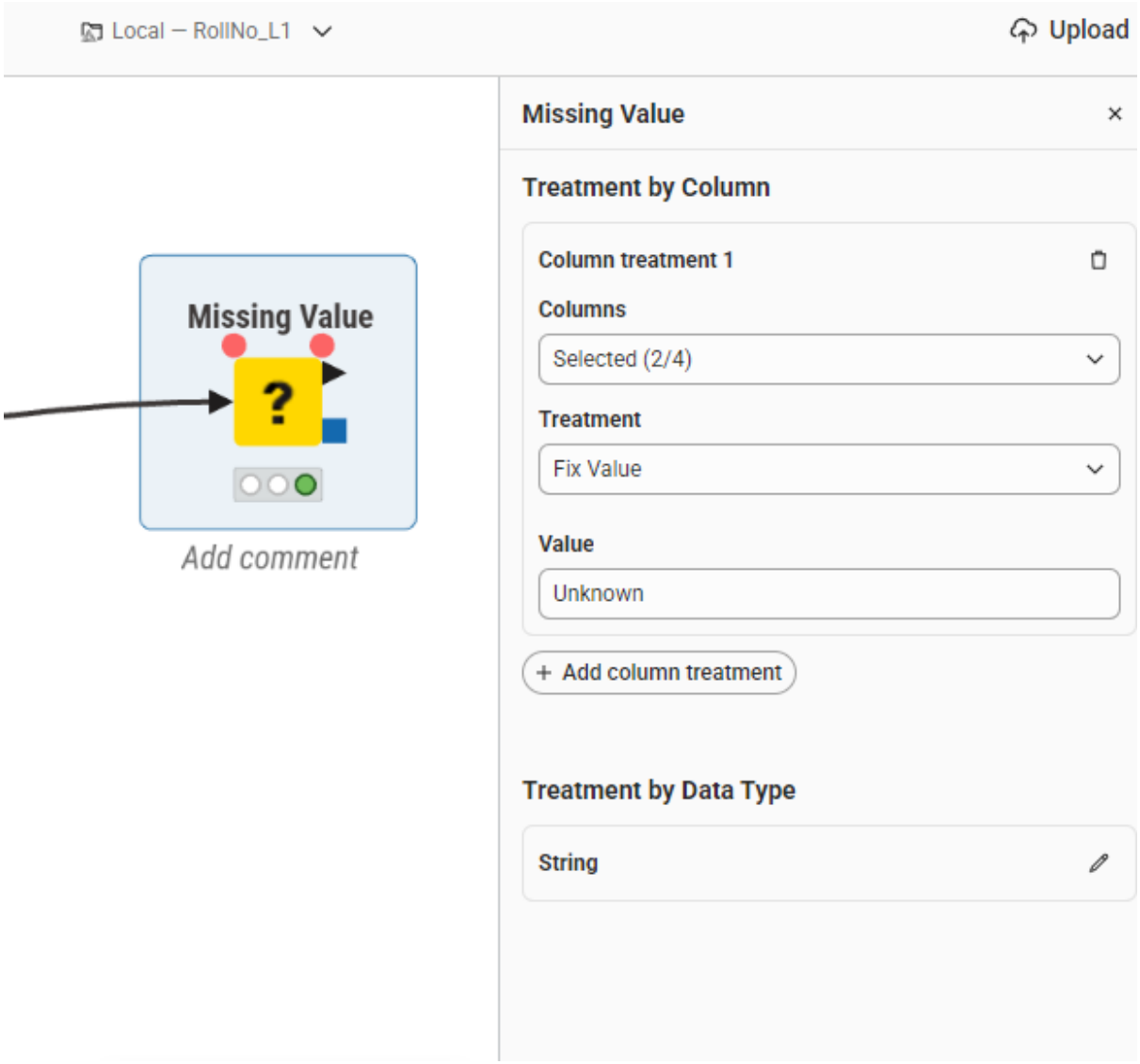


1: File Table | Flow Variables

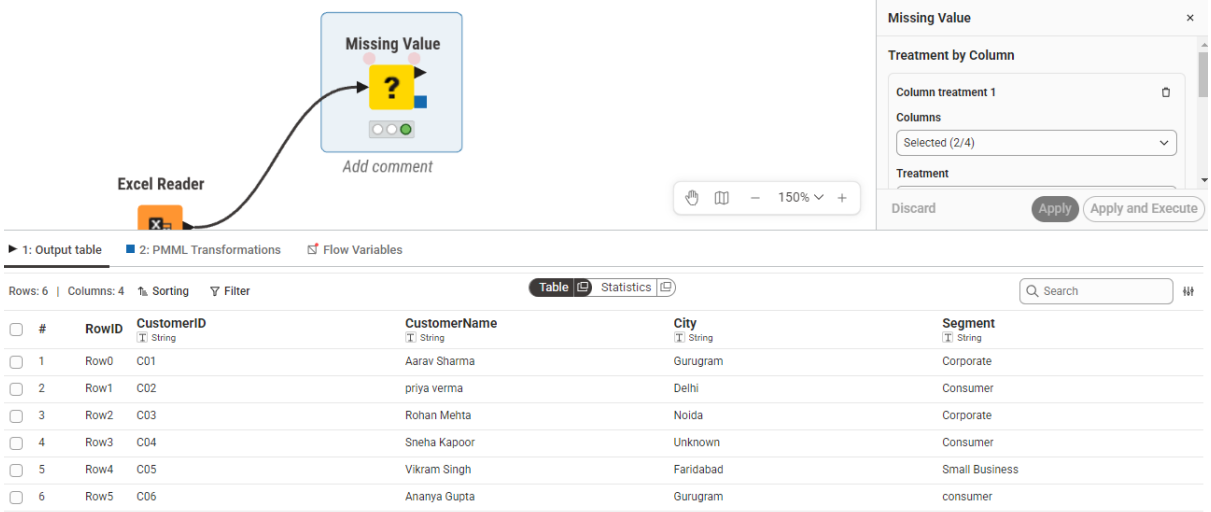
ws: 6 | Columns: 4 | Sorting | Filter | Table | Statistics | Search

#	RowID	CustomerID (String)	CustomerName (String)	City (String)	Segment (String)
1	Row0	C01	Aarav Sharma	Gurugram	Corporate
2	Row1	C02	priya verma	Delhi	Consumer
3	Row2	C03	Rohan Mehta	Noida	Corporate
4	Row3	C04	Sneha Kapoor	?	Consumer
5	Row4	C05	Vikram Singh	Faridabad	Small Business
6	Row5	C06	Ananya Gupta	Gurugram	consumer

Step 29. Drag Missing Value node and connect it to Excel Reader.

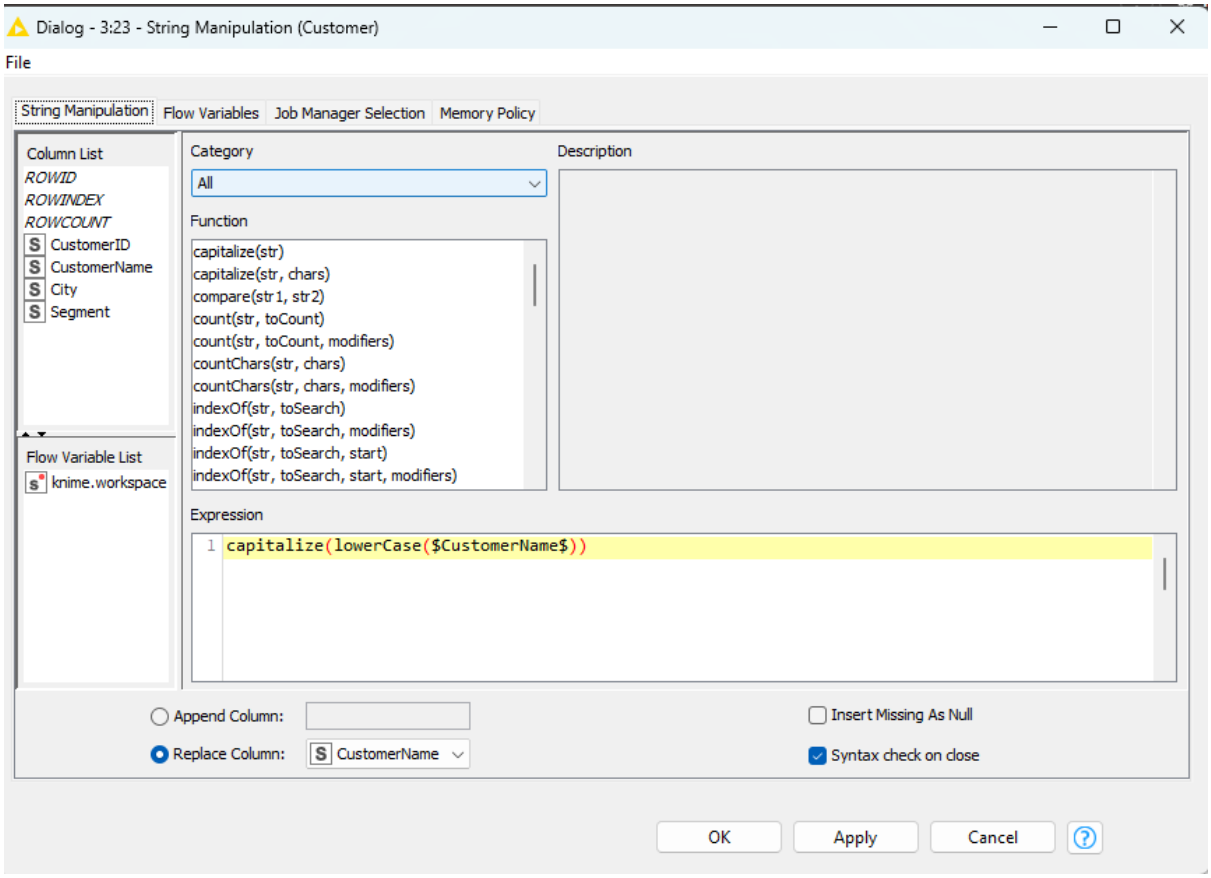


Step 30. Add Treatment and then Execute it.

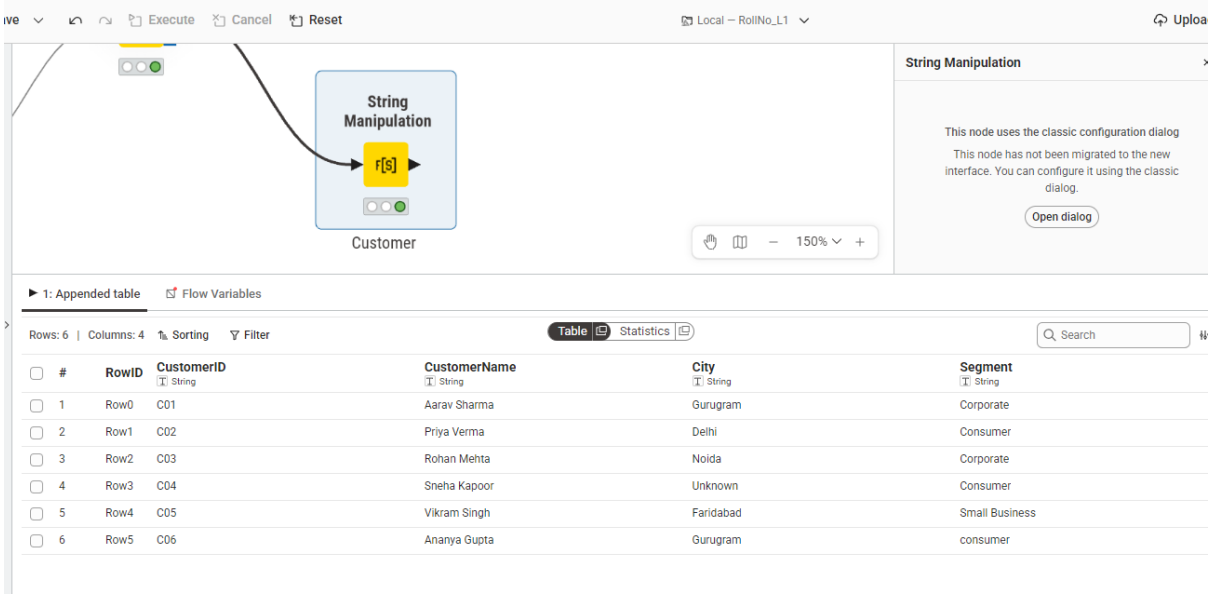


#	RowID	CustomerID	CustomerName	City	Segment
		String	String	String	String
1	Row0	C01	Aarav Sharma	Gurugram	Corporate
2	Row1	C02	priya verma	Delhi	Consumer
3	Row2	C03	Rohan Mehta	Noida	Corporate
4	Row3	C04	Sneha Kapoor	Unknown	Consumer
5	Row4	C05	Vikram Singh	Faridabad	Small Business
6	Row5	C06	Ananya Gupta	Gurugram	consumer

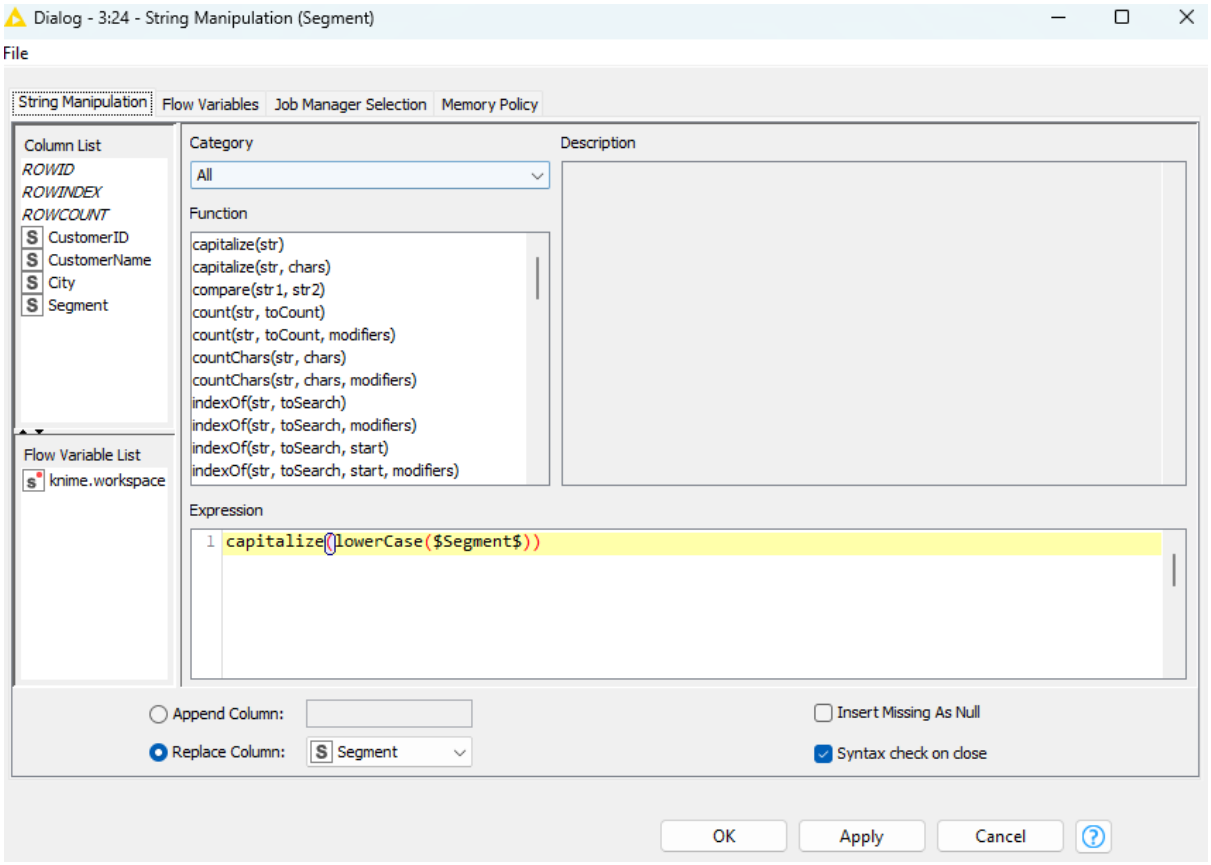
Step 31. Drag String Manipulation node, double click on it and write under Expression.



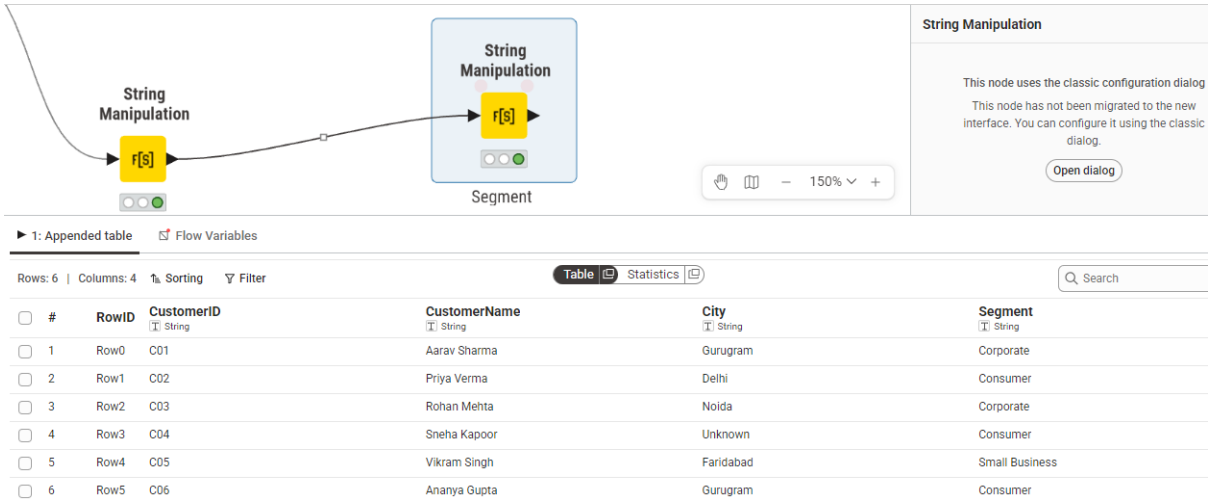
Step 32. Click OK and Execute.



Step 33. Drag another String Manipulation node, double click on it and write:

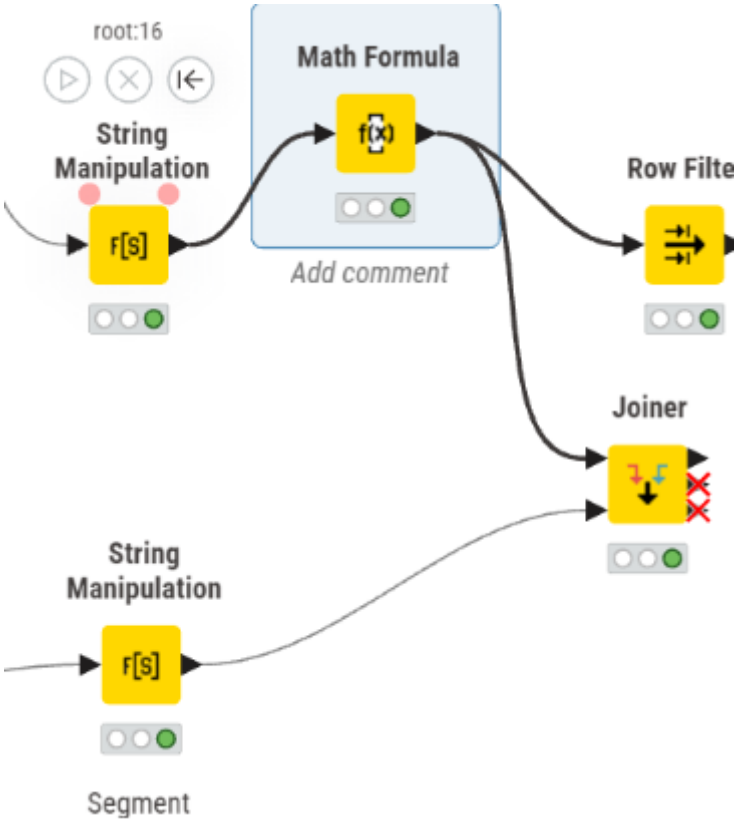


Step 34. Execute String Manipulation.

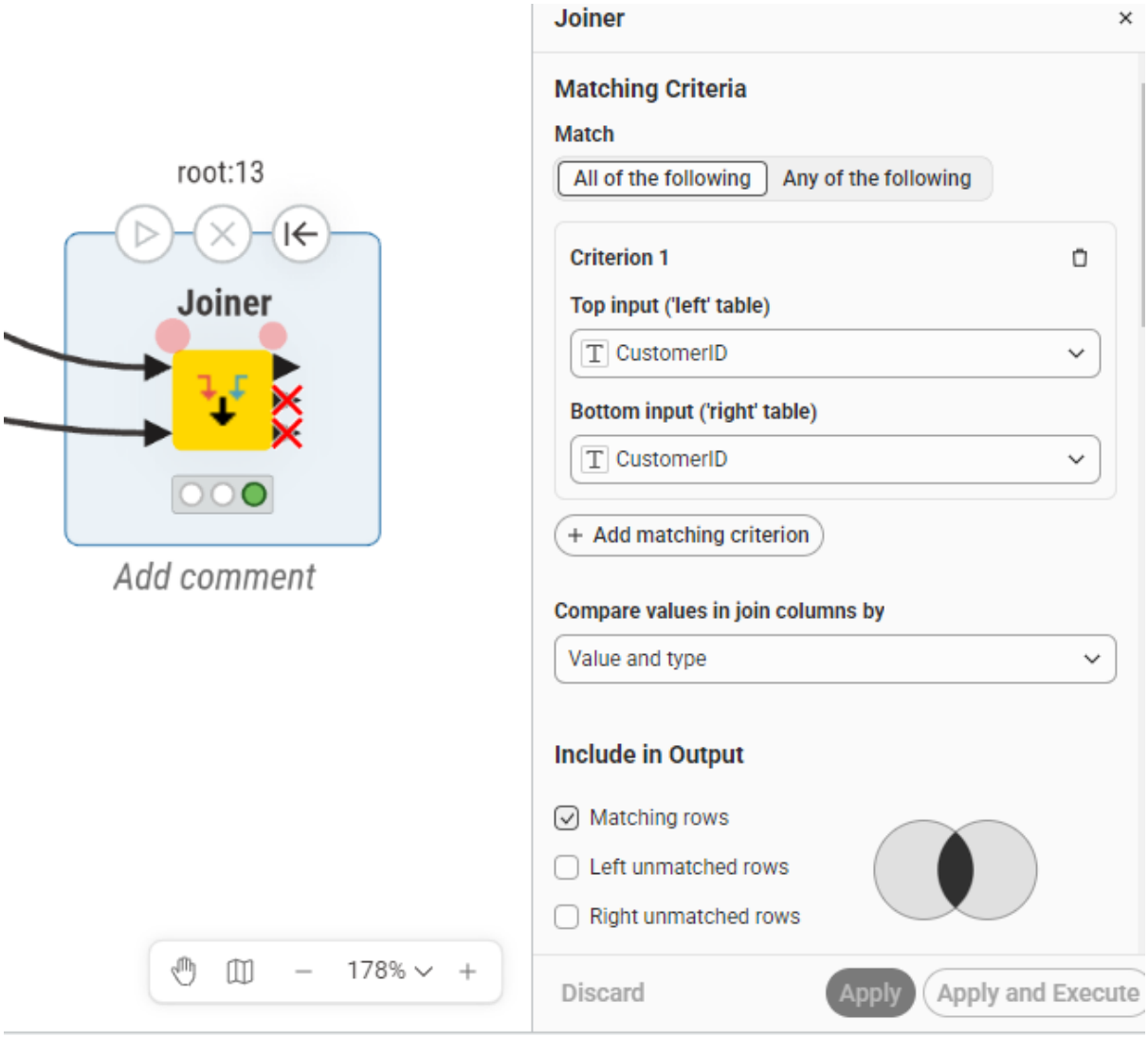


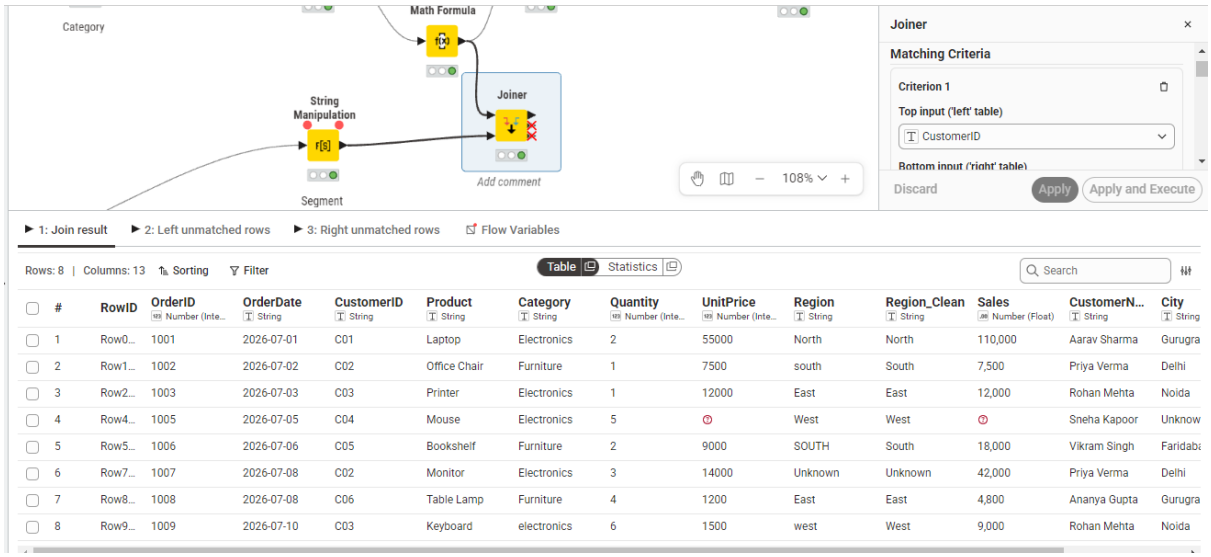
#	RowID	CustomerID (String)	CustomerName (String)	City (String)	Segment (String)
1	Row0	C01	Aarav Sharma	Gurugram	Corporate
2	Row1	C02	Priya Verma	Delhi	Consumer
3	Row2	C03	Rohan Mehta	Noida	Corporate
4	Row3	C04	Sneha Kapoor	Unknown	Consumer
5	Row4	C05	Vikram Singh	Faridabad	Small Business
6	Row5	C06	Ananya Gupta	Gurugram	Consumer

Step 35. Drag a Joiner and connect it to Math Formula (Orders.csv) and String Manipulation (Customer.xlsx).



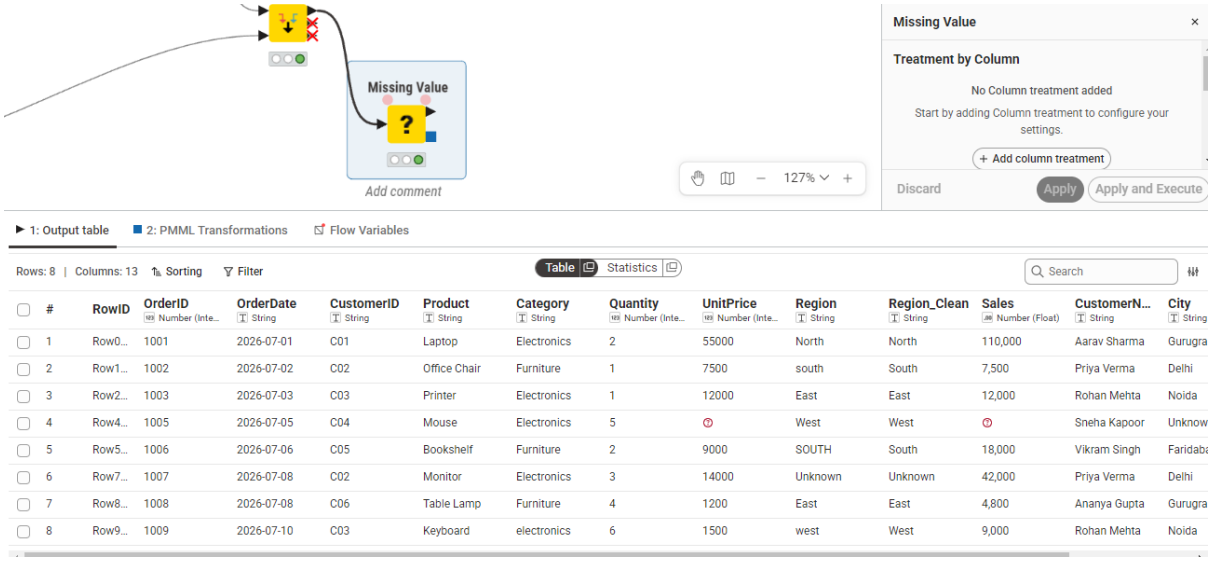
Step 36. Add configuration. Execute.



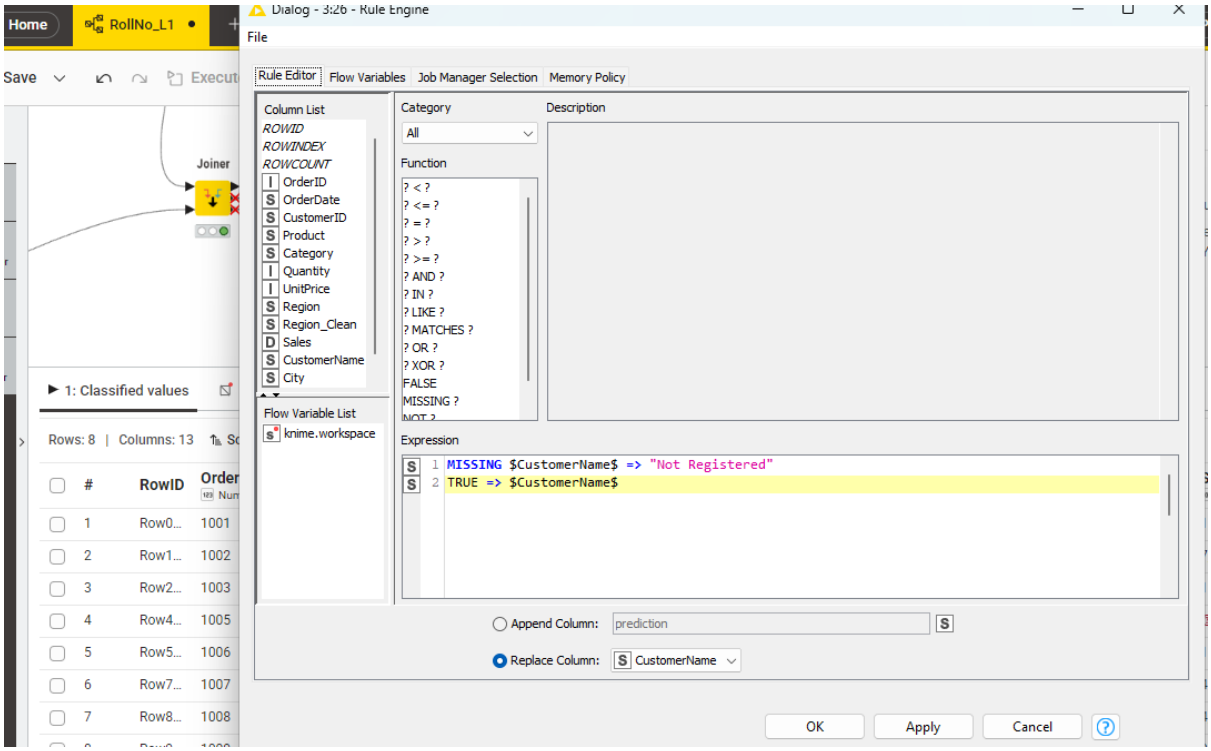


#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Region_Clean	Sales	CustomerN...	City
1	Row0...	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North	110,000	Aarav Sharma	Gurugra
2	Row1...	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South	7,500	Priya Verma	Delhi
3	Row2...	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East	12,000	Rohan Mehta	Noida
4	Row4...	1005	2026-07-05	C04	Mouse	Electronics	5		West	West		Sneha Kapoor	Unknow
5	Row5...	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South	18,000	Vikram Singh	Faridab
6	Row7...	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown	42,000	Priya Verma	Delhi
7	Row8...	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East	4,800	Ananya Gupta	Gurugra
8	Row9...	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West	9,000	Rohan Mehta	Noida

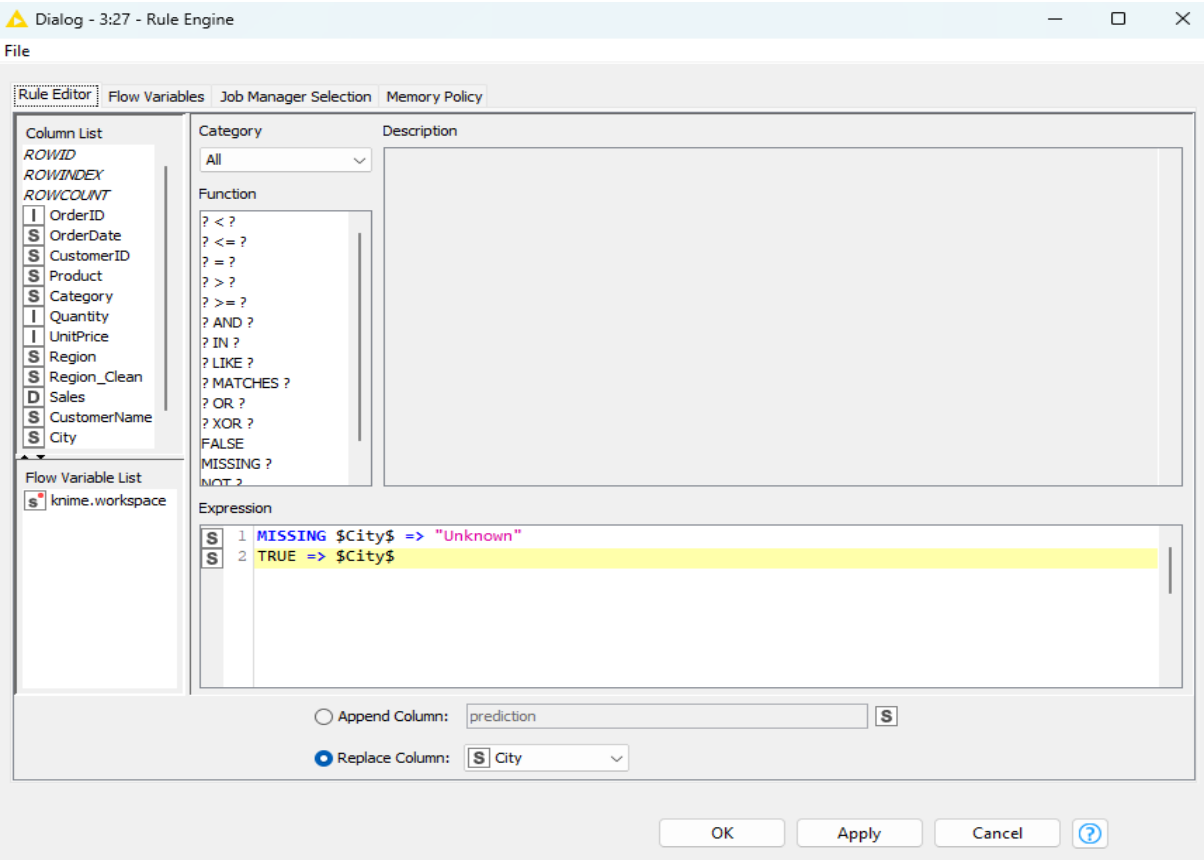
Step 37. Missing Value node, connect it to Joiner.



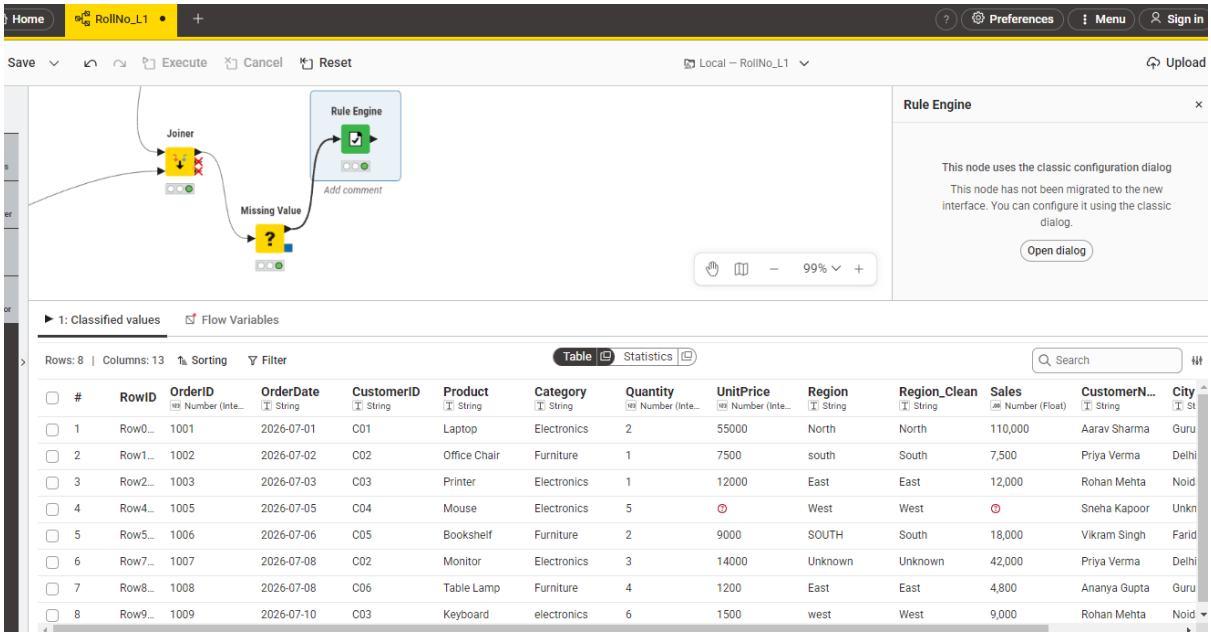
Step 38. Drag Rule Engine node, double click on it and write:



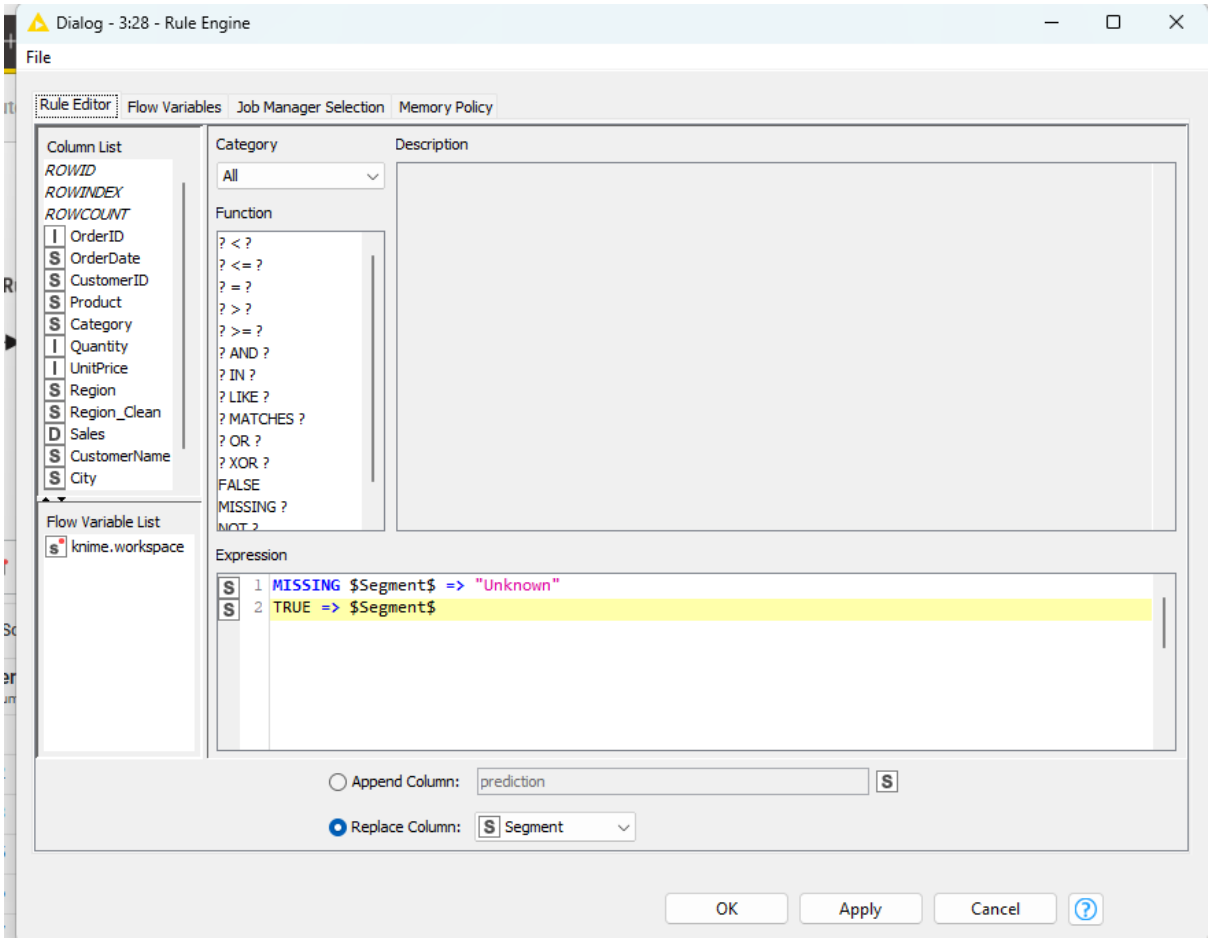
Step 39. Execute Rule Engine. For City, write:



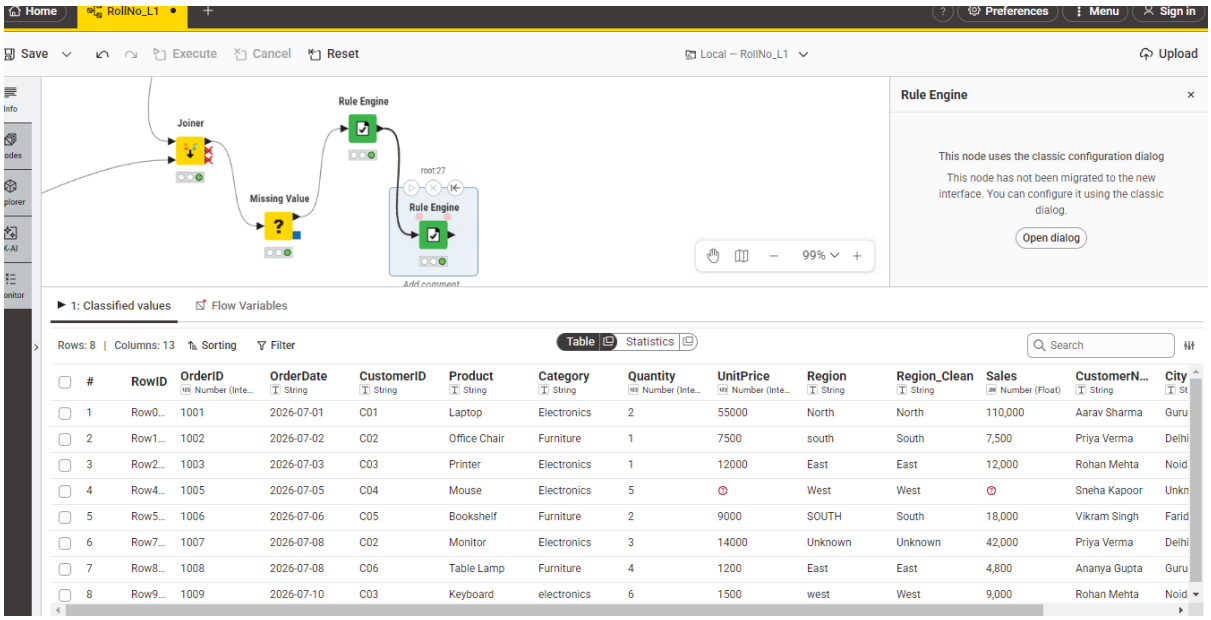
Step 40. Execute.



Step 41. Drag another Rule Engine for Segment.

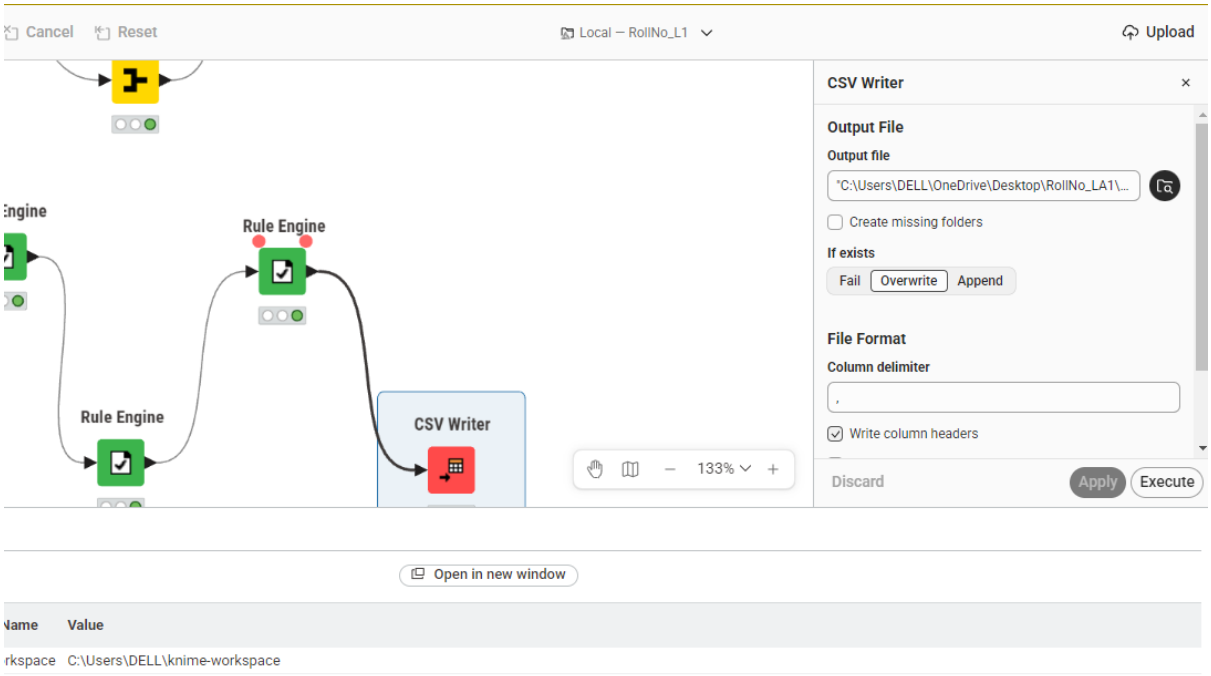


Step 42. Execute.



#	RowID	OrderID	OrderDate	CustomerID	Product	Category	Quantity	UnitPrice	Region	Region_Clean	Sales	CustomerN...	City
1	Row0...	1001	2026-07-01	C01	Laptop	Electronics	2	55000	North	North	110,000	Aarav Sharma	Guru
2	Row1...	1002	2026-07-02	C02	Office Chair	Furniture	1	7500	south	South	7,500	Priya Verma	Delhi
3	Row2...	1003	2026-07-03	C03	Printer	Electronics	1	12000	East	East	12,000	Rohan Mehta	Noid
4	Row4...	1005	2026-07-05	C04	Mouse	Electronics	5	?	West	West	?	Sneha Kapoor	Unkn
5	Row5...	1006	2026-07-06	C05	Bookshelf	Furniture	2	9000	SOUTH	South	18,000	Vikram Singh	Farid
6	Row7...	1007	2026-07-08	C02	Monitor	Electronics	3	14000	Unknown	Unknown	42,000	Priya Verma	Delhi
7	Row8...	1008	2026-07-08	C06	Table Lamp	Furniture	4	1200	East	East	4,800	Ananya Gupta	Guru
8	Row9...	1009	2026-07-10	C03	Keyboard	electronics	6	1500	west	West	9,000	Rohan Mehta	Noid

Step 43. Search and Drag CSV Writer node and connect it to latest Rule Engine. Browse the file name Cleaned_Sales. And Execute it.



The Final Screen:-

